

Matrix-Aware Optimizers: A Survey

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Abstract

Matrix-aware optimizers exploit the row-column structure of matrix-valued neural-network parameters to transform gradients and momentum. This survey examines Shampoo, SOAP, and Muon through their statistical state, update geometry, numerical computation, and distributed implementation. We derive the conditions under which their updates coincide, distinguish factorized bases from Kronecker-product preconditioners, and compare convergence results by their assumptions and stationarity measures. A stochastic-quadratic model shows how preconditioning affects both finite-time contraction and noise accumulation. We also examine inverse-power and polar computations, state transport under basis changes, and matrix updates across shards. Published experiments illustrate the trade-offs between iteration efficiency, numerical accuracy, and runtime. The remaining challenge is to predict when exploiting matrix structure reduces time to a target quality enough to justify its memory and communication costs.

Keywords: adaptive optimization; Shampoo; Muon; Kronecker structure; convergence theory; matrix functions; distributed optimization

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1 Introduction

Matrix-aware optimizers use the row–column structure of parameter blocks to shape their updates. Their appeal is practical: better updates may reduce the cost of training large models. The additional matrix operations, state, and communication can also outweigh any reduction in gradient steps. The relevant comparison is therefore the quality reached within a given training budget.

AdamW combines coordinatewise adaptation with decoupled weight decay, retaining first moments and elementwise second moments [1, 2]. Shampoo, SOAP, and Muon use matrix structure in different ways. Shampoo constructs preconditioners from historical tensor-mode Gram statistics [3, 4]. SOAP learns a factorized basis and applies Adam-style scaling in that basis [5]. Muon applies an approximate polar map to a momentum-based matrix, modifying its singular values [6, 7]. These choices differ in the information they retain, the geometry they impose, and their computational cost.

Published language-model studies report improvements over AdamW in iteration counts and, under particular training recipes, in wall-clock time or training FLOPs [5, 8]. These measures need separate interpretation: a method that needs fewer steps need not reach a target loss sooner. Nor do these results imply a universal ordering among optimizers.

We examine four aspects of each method: its statistical state, update geometry, numerical computation, and distributed representation. This separation helps distinguish changes to the estimator or update from implementations that compute the same update more efficiently. It also clarifies several common ambiguities. Gradient second moments need not equal curvature; a finite polynomial need not approximate polar uniformly; and a factorized basis need not yield a Kronecker-product operator.

The mathematical comparison begins with a common update specification and the conditions under which different maps coincide (sections 4.5 and 5). A frozen stochastic-quadratic model then relates preconditioning to finite-time contraction and noise accumulation. The convergence results are compared by their geometry, oracle, schedule, and controlled quantity (section 6 and appendix B). Numerical and distributed implementations are examined against the update they compute (sections 7 and 8). The explanatory propositions derive these relationships; they do not propose new optimizers or convergence rates.

Related work and scope. The review builds on norm-based and adapted-polar interpretations, KL-based factor fitting, and rotation-based optimization [7, 9–13]. Tables 8 and 9 summarize these connections and their historical context.

Sources were selected for distinct estimation objectives, update maps, proof techniques, numerical methods, or distributed implementations. Broader surveys cover more optimizer families [14]; this review does not attempt an exhaustive treatment of quasi-Newton, natural-gradient, decentralized, or low-memory adaptive methods. K-FAC and Lie-group preconditioning are discussed as related approaches [15, 16]. The appendices contain proof details, theorem specifications, method comparisons, and the conditions of the cited experiments.

2 Mathematical foundations

2.1 Matrix geometry and stochastic information

Let $W \in \mathbb{R}^{m \times n}$ be a parameter block, G_t its stochastic gradient, and F the objective, possibly coupling many such blocks. We use the Frobenius pairing $\langle A, B \rangle = \text{tr}(A^\top B)$ and *column-major* vectorization. Consequently,

$$\text{vec}(AGB) = (B^\top \otimes A) \text{vec}(G) = (B^\top \otimes A)g. \quad (1)$$

For $A \succeq 0$, powers and pseudoinverse powers are defined spectrally. Unless specified otherwise, negative powers require positive definiteness; support-restricted identities set the inverse power to zero on the nullspace.

For a rank- r matrix $Z = U_r \Sigma_r V_r^\top$, its canonical polar factor is

$$\text{polar}(Z) = U_r V_r^\top, \quad \text{polar}(0) = 0. \quad (2)$$

It is a partial isometry, with $\|\text{polar}(Z)\|_F^2 = r$ and spectral norm one when $Z \neq 0$. At rank deficiency we do not complete the nullspace to an arbitrary orthogonal factor. The Frobenius, spectral, and nuclear norms satisfy

$$\|Z\|_F \leq \|Z\|_* \leq \sqrt{r} \|Z\|_F, \quad |\langle A, B \rangle| \leq \|A\|_* \|B\|_{\text{op}}. \quad (3)$$

The nuclear norm is dual to the spectral norm; the chosen norm determines the descent direction and stationarity measure.

Let filtration $\mathcal{F}_{t-1}$ contain the history before the current sample, with W_t measurable with respect to that history. A conventional unbiased oracle satisfies

$$\mathbb{E}[G_t | \mathcal{F}_{t-1}] = \nabla F(W_t), \quad N_t = G_t - \nabla F(W_t). \quad (4)$$

A preconditioner formed from G_t is generally not $\mathcal{F}_{t-1}$ -measurable. Thus unbiasedness does not imply $\mathbb{E}[P_t \text{vec}(N_t) | \mathcal{F}_{t-1}] = 0$. Analyses of current-gradient-dependent adaptation must control that dependence or use a suitably delayed construction.

2.2 Second moment, covariance, Fisher, and Hessian

At a fixed parameter, write $g = \text{vec}(G)$, $\mu = \mathbb{E}[g]$. The uncentered second moment and centered noise covariance are

$$C = \mathbb{E}[gg^\top] = \Sigma + \mu\mu^\top, \quad \Sigma = \text{Cov}(g). \quad (5)$$

For a mean of B conditionally independent example gradients,

$$C_B = \mu\mu^\top + \frac{\Sigma_{\text{sample}}}{B}. \quad (6)$$

A Gram statistic of the minibatch mean thus weights the mean and noise differently from an average of per-example Gram statistics.

The Hessian $H = \nabla^2 F$ describes local curvature and can be indefinite. A model Fisher averages score outer products under the model distribution. Equating it with a Hessian requires a likelihood

model and regularity conditions; equating it with a gradient second moment also depends on the sampling distribution. K-FAC approximates curvature/Fisher using model structure [15], whereas Shampoo accumulates gradient Gram matrices [3]. Interpreting the latter as covariance estimates requires additional assumptions.

The population mode contractions

$$L = \mathbb{E}[GG^\top], \quad R = \mathbb{E}[G^\top G] \quad (7)$$

are partial traces of C . They retain row-column information, but matrices with the same contractions can have different cross-mode correlations. Hence they do not generally determine all $mn \times mn$ entries of C .

2.3 AdamW as a baseline and an algorithmic specification

With elementwise multiplication and division understood, the AdamW template is

$$\begin{aligned} M_t &= \beta_1 M_{t-1} + (1 - \beta_1) G_t, \\ V_t &= \beta_2 V_{t-1} + (1 - \beta_2) G_t^{\odot 2}, \\ W_{t+1} &= (1 - \eta_t \lambda_t) W_t - \eta_t \frac{\widehat{M}_t}{\sqrt{\widehat{V}_t + \varepsilon}}. \end{aligned} \quad (8)$$

Hats denote the usual bias corrections. The decoupled decay in (8) is not generally the update obtained by inserting an ℓ_2 penalty into an adaptively preconditioned gradient [2]. Diagonal adaptation omits cross-coordinate second moments, making its updates inexpensive and separable across shards.

A useful common form is

$$S_t = \mathcal{A}(S_{t-1}, G_t), \quad \widehat{D}_t = \mathcal{T}(S_t, Z_t), \quad W_{t+1} = (1 - \eta_t \lambda_t) W_t - \eta_t \widehat{D}_t, \quad (9)$$

where S_t is statistical or numerical state and Z_t may be a raw gradient, momentum, or corrected estimator. The map $\mathcal{T}$ can apply a preconditioner, transform singular values, or rescale its output. Even when the transformation is linear for fixed S_t , its dependence on the current gradient through S_t can make the full gradient-to-update map nonlinear.

Operator and sign conventions. An *update preconditioner* acts directly on the vectorized gradient or momentum. When a proof instead uses a metric H_t , the acting preconditioner is H_t^{-1} ; the moment matrices from which it is constructed are separate objects. We reserve D_t for a direction subtracted by the update, as in (9), and use Δ for a signed trial displacement in linear-minimization problems.

Specifying the algorithm requires the inputs to S_t and $\mathcal{T}$, the averaging rule, damping placement, and refresh schedule. Replacing G_t by M_t in a Gram estimate, or M_t by a Nesterov combination in a polar map, changes the algorithm.

2.4 A factorized basis need not imply a factorized operator

For SPD factors, $D = L^{-a} Z R^{-b}$ corresponds to $\text{vec}(D) = (R^{-b} \otimes L^{-a}) \text{vec}(Z)$. In contrast, coordinate scaling in a factorized orthogonal basis has the form

$$P = Q \text{diag}(\text{vec}(s)) Q^\top, \quad Q = Q_R \otimes Q_L, \quad (10)$$

with an arbitrary positive $m \times n$ array s . A single Kronecker operator in that basis requires $s_{ij} = a_i b_j$. The array $(\frac{1}{1} \frac{1}{2})$ fails this condition. SOAP therefore has a Kronecker-factored basis without necessarily having separable adaptive scales [5, 10].

Rotation as an update coordinate system. For orthogonal U_t, V_t , a broad template is

$$\tilde{Z}_t = U_t^\top Z_t V_t, \quad D_t = U_t \mathcal{H}_t(\tilde{Z}_t) V_t^\top. \quad (11)$$

The basis and the rule $\mathcal{H}_t$ are independent choices. Inverse powers give separable scales in a factor eigenbasis; SOAP uses entrywise adaptive scales. In the current input’s singular-vector basis, polar and finite-polynomial rules specify a singular-value response. ARO combines adaptive rotation with normed steepest descent [12]. Convergence results for rotation-based methods apply to particular state and normalization recursions [13], detailed in [section 6.4](#) and [appendix B.10](#).

Table 1: Design principles of matrix-aware optimization. Methods may combine several rows.

Principle	State retained	Transformation	Principal theoretical question
Coordinate adaptation	Entrywise moments	Diagonal scaling	Which coordinate geometry is favorable?
Mode adaptation	Mode Gram or fitted factors	One-/two-sided inverse powers	Which structured statistic and exponent are justified?
Basis adaptation	Mode bases and coordinate moments	Rotated diagonal scaling	How are scales and historical state transported?
Spectral update	Momentum or corrected gradient	Polar or finite polynomial	Which norm geometry and alignment are controlled?
Adaptive spectral update	Spectral input plus adaptive state	Pre/post scaling and orthogonalization	Does composition preserve a descent argument?
Compressed update	Subspaces, residuals, or low-rank state	Restricted spectral transformation	What information and convergence properties survive?

3 Structured adaptive preconditioning

3.1 From full-matrix adaptation to mode statistics

Full-matrix AdaGrad uses a cumulative outer-product matrix whose dimension equals the number of parameters [17]. Shampoo replaces it with tensor-structured statistics. For a matrix block, the canonical cumulative update is

$$\begin{aligned} L_t &= \varepsilon I_m + \sum_{s=1}^t G_s G_s^\top, \\ R_t &= \varepsilon I_n + \sum_{s=1}^t G_s^\top G_s, \\ W_{t+1} &= W_t - \eta_t L_t^{-1/4} G_t R_t^{-1/4}. \end{aligned} \tag{12}$$

The vectorized metric is $H_t = R_t^{1/4} \otimes L_t^{1/4}$, with update $-\eta_t H_t^{-1} g_t$. In an order- k tensor, one factor is formed for each mode and the original construction uses exponent $-1/(2k)$ per mode [3]. This preserves tensor structure without storing a full parameter-space second moment.

The original regret analysis compares the cumulative vectorized outer product with the product of mode contractions, yielding a bound that depends on traces of fourth roots and gradient rank. The derivation in [section 6.2](#) and [appendix A.1](#) explains the fourth roots in the original algorithm, although other whitening objectives can lead to different exponents.

One-sided variants. One-sided updates such as $L_t^{-1/2} G_t$ can be attractive for highly rectangular blocks, but change the operator class, exponent, and proof geometry. ASGO and unified structured-preconditioner analyses treat them separately [18, 19]. Comparing dimension dependence also requires matching the radius and smoothness assumptions in the bounds.

3.2 Kronecker approximation and whitening

Suppose, as a population model, that $C = B \otimes A$ with $A, B \succ 0$. Then the contractions in (7) satisfy

$$L = \text{tr}(B)A, \quad R = \text{tr}(A)B, \quad s = \mathbb{E} \|G\|_F^2 = \text{tr}(A) \text{tr}(B). \tag{13}$$

Consequently,

$$C = \frac{R \otimes L}{s}, \quad \text{unvec}\left(C^{-1/2} \text{vec}(G)\right) = \sqrt{s} L^{-1/2} G R^{-1/2}. \tag{14}$$

These Kronecker and trace identities show that inverse fourth roots of the *same* Gram matrices do not generally give exact whitening by $C^{-1/2}$.

Morwani et al. [20] relate Shampoo's factors to an approximation procedure for the gradient second-moment matrix, giving factor estimation a statistical objective beyond storage savings.

3.3 SOAP: separate the basis from the scales

Write eigendecompositions $L_t = Q_{L,t}\Lambda_{L,t}Q_{L,t}^\top$ and $R_t = Q_{R,t}\Lambda_{R,t}Q_{R,t}^\top$. Within an interval where the bases are fixed, SOAP rotates gradients into this basis and applies coordinatewise Adam-style adaptation [5]:

$$\begin{aligned}\tilde{G}_t &= Q_L^\top G_t Q_R, \\ \tilde{M}_t &= \beta_1 \tilde{M}_{t-1} + (1 - \beta_1) \tilde{G}_t, \\ \tilde{V}_t &= \beta_2 \tilde{V}_{t-1} + (1 - \beta_2) \tilde{G}_t^{\odot 2}, \\ D_t &= Q_L \left(\frac{\tilde{M}_t}{\sqrt{\tilde{V}_t} + \varepsilon} \right) Q_R^\top.\end{aligned}\tag{15}$$

SOAP can refresh its basis less frequently than its coordinatewise moments. The dense array $\tilde{V}_t$ gives the larger operator class in (10), allowing scales that are not separable across the two modes.

Basis refreshes also require a rule for the moment state. A full first moment can be stored in the original coordinates and projected into the current basis. In Algorithm 3 of the inspected SOAP paper, the diagonal array continues its entrywise EMA across refreshes, mixing observations expressed in successive bases rather than reconstructing the historical second moment [5, Section 4.1]. Permuting, resetting, or approximately rotating this state defines a different rule.

A diagonal state generally lacks the information required for exact rotated diagonals (proposition 7.4). Yet Sun et al. [13] establish convergence for a specified moving-basis EMA recurrence, including rotations that depend on the current gradient. Their normalizer is $\sqrt{\tilde{V}_t} + \varepsilon$ rather than $\sqrt{\tilde{V}_t} + \varepsilon$, and they retain the first moment in original coordinates. The recursion and schedules in section 6.4 and appendix B.10 differ from the fixed-basis template above.

Eigenvalue correction and grafting. Learning-rate grafting replaces D_t by

$$D_t^{\text{graft}} = \frac{\|D_t^{\text{base}}\|_F}{\|D_t\|_F} D_t\tag{16}$$

when the denominator is nonzero, with an implementation-specific zero rule. It preserves a block's direction and imports another method's magnitude. Eschenhagen et al. [21] disentangle this effect from stale eigenbases and eigenvalue correction. Their analysis motivates examining the update spectrum.

3.4 KL-based fitting and the statistical meaning of the factors

Factors can instead be fitted through a separable Gaussian approximation. For a positive-definite target second moment C and approximation $B \otimes A$, the zero-mean Gaussian divergence $D_{\text{KL}}(\mathcal{N}(0, C) \parallel \mathcal{N}(0, B \otimes A))$, up to constants and a factor $1/2$, is

$$J(A, B) = n \log \det A + m \log \det B + \text{tr}[(B^{-1} \otimes A^{-1})C].\tag{17}$$

Its block stationary conditions are

$$A = \frac{1}{n} \mathbb{E}[GB^{-1}G^\top], \quad B = \frac{1}{m} \mathbb{E}[G^\top A^{-1}G].\tag{18}$$

Each mode factor is estimated using the other mode’s geometry ([appendix A.2](#)). The gauge transformation $(A, B) \mapsto (cA, B/c)$ leaves the Kronecker product unchanged, so factor norms require a normalization convention.

KL-Shampoo and KL-SOAP use this objective to redesign estimation and basis/scale updates [10]. Treating an uncentered second moment as a zero-mean Gaussian target is a modeling choice; alternative Bregman objectives imply different fitting rules [22].

3.5 Practical variants

Statistical state and factor dynamics. Sketchy uses Frequent Directions sketches of Shampoo’s mode statistics, with an online-convex memory–regret trade-off governed by discarded spectral information [23]. Root-free and inverse-free Shampoo redesign the factor recurrence to avoid explicit roots and inversions [24]; they cannot be obtained by removing the root solver from the original recurrence.

Update rule. SOAP and eigenvalue-corrected variants change the represented operator. SPlus combines a historical eigenbasis with bounded instantaneous normalization, shape-aware scaling, and iterate averaging [25], changing the update’s magnitude and averaging as well as its basis.

Numerical state and execution. Scalable and distributed Shampoo use blocking, delayed expensive operations, and systems choices [4, 26]. Blocking can change the geometry, while delayed computation changes which numerical state is current. FOAM relates damping to refresh frequency and operator error [27]. Reparametrized subspace updates and bfloat16 storage change the numerical representation of the state [28], while DASH improves block-preconditioning and inverse-root execution [29].

Four-bit Shampoo quantizes the eigenbasis and corrects its loss of orthogonality in the inverse-root preconditioner [30]. Unlike Sketchy’s compression of the statistic, this introduces numerical representation error rather than rank truncation; the four-bit designation does not cover every state or arithmetic operation.

[Section 6](#) compares the estimator, regret, descent, and numerical guarantees available for these recursions.

4 Spectral and adaptive spectral updates

4.1 Idealized Muon and its norm geometry

An idealized Muon step applies a polar transformation to a momentum-like matrix Z_t :

$$D_t = a_t \text{polar}(Z_t), \quad W_{t+1} = (1 - \eta_t \lambda_t) W_t - \eta_t D_t. \quad (19)$$

Here a_t includes a block or shape scale. Orthogonalization acts on the *update input*, not the parameter matrix W_t , so Muon does not generally constrain weights to a Stiefel manifold. For rank-deficient Z_t , the canonical transformation is a partial isometry rather than a full-rank orthogonal completion.

The norm-based interpretation follows from a linear minimization problem [6, 7]:

$$\min_{\|\Delta\|_{\text{op}} \leq \rho} \langle G, \Delta \rangle = -\rho \|G\|_*, \quad \Delta^* = -\rho \text{polar}(G). \quad (20)$$

Duality gives the lower bound, attained by the compact SVD. This defines steepest linearized descent within a spectral-norm ball, whereas

$$\arg \min_{\Delta} \left\{ \langle G, \Delta \rangle + \frac{1}{2\eta} \|\Delta\|_{\text{op}}^2 \right\} \ni -\eta \|G\|_* \text{polar}(G). \quad (21)$$

The two problems select the same polar direction with different magnitudes. Normalized, squared-norm, and preconditioned gradient descent thus specify different updates.

Momentum conventions. A two-coefficient momentum template is

$$M_t = \beta_1 M_{t-1} + (1 - \beta_1) G_t, \quad Z_t = \beta_2 M_t + (1 - \beta_2) G_t. \quad (22)$$

This gives a heavy-ball input at $\beta_2 = 1$ and a common normalized Nesterov-style combination at $\beta_2 = \beta_1$. Implementations also use the unnormalized buffer $B_t = \beta B_{t-1} + G_t$. With matching initialization and fixed coefficients, the inputs may differ by a positive scalar removed by exact polar. Normalization floors, varying coefficients, or adaptive post-scaling can break this equivalence.

4.2 The actual finite-polynomial update

Muon uses fast approximate orthogonalization to avoid repeated SVD [6]. A representative fifth-degree iteration is

$$X_0 = Z/\tau, \quad X_{j+1} = a_j X_j + b_j X_j X_j^\top X_j + c_j (X_j X_j^\top)^2 X_j. \quad (23)$$

The multiplication order may use the smaller Gram matrix. In exact arithmetic and for a fixed input SVD,

$$X_K = U \text{diag}(q_K(\sigma_i/\tau)) V^\top, \quad q_K = \phi_{K-1} \circ \cdots \circ \phi_0, \quad (24)$$

where $\phi_j(s) = a_j s + b_j s^3 + c_j s^5$. The finite iteration defines a spectral filter. The original empirical triple is $(3.4445, -4.7750, 2.0315)$, often used for five steps [6, 31]. Since $\phi(1) = 0.701$, increasing the step count does not turn it into a classical fixed-point iteration converging to one.

Taylor-family NS analyses control convergence toward a polar factor under spectral conditions [32], while finite-map analyses control the output's alignment and energy [31, 33, 34]. A finite map may give a useful optimization direction despite a large worst-case polar error. Accurate polar computation alone does not resolve momentum tracking error, stochastic noise, or an unsuitable learning-rate schedule.

Polar Express develops optimized matrix-sign/polar procedures and finite-precision analysis [35]. Outer-optimizer convergence additionally requires assumptions connecting the numerical output to the gradient process.

4.3 Scale, parameter classes, and adaptation

For a full-rank $m \times n$ input, let $r = \min(m, n)$. Then

$$\text{RMS}(\text{polar}(Z)) = \frac{\|\text{polar}(Z)\|_F}{\sqrt{mn}} = \frac{1}{\sqrt{\max(m, n)}}. \quad (25)$$

A factor proportional to $\sqrt{\max(m, n)}$ matches entrywise RMS across full-rank matrix shapes. Rank deficiency, finite polynomials, or a different target update distribution change this calculation. Scalable Muon revisits scaling and weight decay for large language models [8]; learning rates can only be compared together with their update scales.

Scalar output adaptation. AdaGO sets the magnitude using clipped current and accumulated gradient norms while preserving the polar direction [36]. Its scalar adaptation changes neither singular directions nor relative singular-value magnitudes.

Input adaptation. Variance-Adaptive Muon’s Muon-NSR and Muon-VS modulate momentum using variance information before orthogonalization [37]. Muon² applies Adam-style second-moment preconditioning at the same stage; Muon²-F factorizes the additional state [38]. Both the spectral-map input and its statistical memory change.

Spectral response and coordinate-dependent output. BeyondMuon studies $U\Sigma^pV^\top$ transformations of momentum or RMS-normalized inputs [39]. The exponent controls retained singular-value contrast; the study does not establish a universally best choice. AdaMuon combines elementwise second-moment adaptation of orthogonalized directions with, in its inspected v3, a sign transformation of momentum before orthogonalization [40]. NorMuon uses neuron-wise normalization/adaptation and a distributed implementation [41]. Coordinate-dependent scaling after polar generally removes its equal-singular-value property, requiring analysis of the composed update.

Curvature-aware compositions. Mousse orthogonalizes in coordinates determined by Kronecker-factored statistics [42]. The weighted spectral-ball construction in [proposition 5.3](#) includes both the coordinate transformation and the map back. In general, $\text{polar}(AZB)$ is not $A\text{polar}(Z)B$: adaptation and orthogonalization do not commute. Convergence must therefore be established for the chosen composition order.

Coverage beyond dense hidden matrices. Muon can handle selected matrices while another optimizer handles embeddings, biases, or normalization parameters. Kimi K2’s technical report describes a stabilization-oriented recipe at much larger scale [43], providing evidence about a complete training system. AF-Muon uses a support-aware finite-cap rule for tied vocabulary tables and RMS normalization for one-dimensional parameters [44]. These choices assign different geometry to different parameter roles within one model.

4.4 Allocating geometry across subspaces, layers, and time

Dion uses low-rank orthonormalized updates with state and communication mechanisms for distributed training [45]; Dion3 extends this approach across the training stack [46]. Their relation to exact Muon depends on how omitted directions are handled—through residuals, tracked subspaces, or a different geometry—and equality requires proof.

COSMOS applies SOAP-like adaptation in a tracked leading subspace and a Muon-style update to its complement [47]. The rank, mixture, and normalization determine both geometry and memory cost.

TEON groups compatible layer gradients into a higher-order tensor and chooses a matricization for orthogonalization [48]. This changes the geometry beyond batching independent matrix operations. LionMuon alternates periodic spectral steps with sign steps, sharing momentum with branch-specific step sizes [49]; its schedule changes both the recurrence and the frequency of orthogonalization.

LiMuon combines recursive variance reduction with randomized-SVD compression of the carried estimator [50]. Its guarantees concern this estimator and a paired, same-sample gradient oracle, whose additional evaluations must be counted in sample-complexity and runtime comparisons.

Early theory analyzes idealized Muon and its stochastic behavior [51]. Later work covers convergent NS families, empirical finite polynomials, Nesterov tracking, and variance-reduced estimators [31–34]. Deterministic regularized proxies admit a Lyapunov analysis [52], while nonconvergence constructions and approximation-error analyses delimit particular settings [53]. Algebraic connections appear in [section 5](#), and guarantees for each recursion and oracle in [section 6](#).

4.5 Comparing the core update rules

[Table 2](#) summarizes four core rules for one $m \times n$ block, describing the gradient-driven direction in (9). Learning rate, output scaling, and decoupled decay are outer choices; state, numerical, and distributed specifications appear in [appendix C](#).

Table 2: Core update rules after fixing the current optimizer state. A fixed-state linear form does not make the full gradient-to-update map linear when that state depends on the current gradient.

Method	Retained information	Representative transformation	Main distinction
AdamW	Entrywise first and second moments	Coordinatewise scaling of momentum, (8)	Scaling in the original coordinates
Shampoo	Historical mode Gram matrices	Two-sided inverse powers, (12)	A separable update operator
SOAP	Mode bases and coordinatewise moments	Rotate, adapt, and rotate back, (15)	Factored basis; generally nonseparable scales
Finite Muon	Momentum with a declared input convention	A fixed-depth singular-value map, (24)	Spectral transformation rather than covariance estimation

5 Algebraic connections and optimization dynamics

5.1 Instantaneous and accumulated adapted-polar identities

Proposition 5.1 (Instantaneous identity and its damped filter). *For any G , with inverse powers restricted to its singular support,*

$$(GG^\top)^{-1/4}G(G^\top G)^{-1/4} = \text{polar}(G). \quad (26)$$

With the same damping $\delta > 0$ on both Gram factors, the singular response becomes

$$(\sigma^2 + \delta)^{-1/4}\sigma(\sigma^2 + \delta)^{-1/4} = \frac{\sigma}{\sqrt{\sigma^2 + \delta}}. \quad (27)$$

Proof. Substitute the compact SVD and cancel the singular-value powers on the support. On the nullspace both sides of (26) are zero. The damped expression follows by applying the scalar function to each singular value. $\square$

This connects instantaneous Shampoo and polar updates for the *same input matrix*, without historical accumulation or a separate momentum choice [6, 7]. Damping changes the response to (27).

Proposition 5.2 (An adapted-polar decomposition). *For SPD L, R , real exponents a, b , and any matrix M ,*

$$L^{-a}MR^{-b} = [L^{-a}(MM^\top)^{1/4}] \text{polar}(M)[(M^\top M)^{1/4}R^{-b}]. \quad (28)$$

Proof. The compact SVD gives $(MM^\top)^{1/4} \text{polar}(M)(M^\top M)^{1/4} = M$. Multiplication by the two outside factors proves the claim without a commutativity assumption. $\square$

The adapted-polar viewpoint of Eschenhagen et al. [9] accommodates accumulated statistics through (28). The bracketed factors need not be symmetric or positive definite. The current singular factors depend on M , while L, R may encode a nonstationary history, so the decomposition alone does not imply identical training trajectories.

For a full-row-rank $G \in \mathbb{R}^{m \times n}$ with $m \leq n$, the short-sided limit is

$$\left(\frac{1}{n}GG^\top\right)^{-1/2}G = \sqrt{n} \text{polar}(G). \quad (29)$$

The short-sided KL connection in the inspected v10 of Lin et al. [10] uses this instantaneous limit, which differs from the population fitting problem and its online solver.

5.2 Weighted spectral geometry and composition order

Proposition 5.3 (Spectral descent in anisotropic coordinates). *Let $A \succ 0$, $B \succ 0$, and $\rho > 0$. A minimizer of*

$$\min_{\Delta} \langle G, \Delta \rangle \quad \text{subject to} \quad \|A^{1/2} \Delta B^{1/2}\|_{\text{op}} \leq \rho \quad (30)$$

is

$$\Delta^* = -\rho A^{-1/2} \text{polar}(A^{-1/2} G B^{-1/2}) B^{-1/2}. \quad (31)$$

Proof. Set $Y = A^{1/2} \Delta B^{1/2}$. The objective is $\langle A^{-1/2} G B^{-1/2}, Y \rangle$, so (20) applies, followed by the inverse coordinate transformation. $\square$

This interprets curvature-aware spectral compositions such as Mousse [42], without establishing that gradient-estimated A and B equal curvature factors. Omitting the output transformation defines a different method.

Equivariance. For orthogonal U, V , $\text{polar}(U Z V^\top) = U \text{polar}(Z) V^\top$. Exact Shampoo has the same left/right orthogonal equivariance with isotropic initialization and consistently transformed history. Arbitrary rotations in the flattened mn -dimensional space need not preserve matrix/Kronecker structure, and coordinatewise scaling generally has less invariance. For basis-dependent methods, repeated eigenspaces make eigenbasis choices nonunique; operator theorems must account for them. Claims of rotation invariance must specify the transformation group and state transformation.

5.3 A frozen quadratic separates curvature from noise

A stochastic quadratic with a fixed preconditioner separates initial-error contraction from noise accumulation:

$$f(x) = \frac{1}{2} x^\top H x, \quad x_{t+1} = x_t - \eta P(H x_t + \xi_t), \quad (32)$$

where $H, P \succ 0$ are fixed. Let x_t be measurable with respect to a filtration $\mathcal{H}_t$, let ξ_t be $\mathcal{H}_{t+1}$ -measurable, and assume $\mathbb{E}[\xi_t | \mathcal{H}_t] = 0$ and $\mathbb{E}[\xi_t \xi_t^\top | \mathcal{H}_t] = \Sigma$ for a fixed finite matrix $\Sigma \succeq 0$. The initial state is deterministic or satisfies $\mathbb{E} \|x_0\|^2 < \infty$. The conditional noise covariance is time independent; temporal independence is not required. Define

$$z = P^{-1/2} x, \quad \tilde{H} = P^{1/2} H P^{1/2}, \quad \tilde{\Sigma} = P^{1/2} \Sigma P^{1/2}. \quad (33)$$

Then $z_{t+1} = (I - \eta \tilde{H}) z_t - \eta P^{1/2} \xi_t$.

Proposition 5.4 (Stability, finite-time loss, and stationary loss). *The deterministic recursion is strictly stable precisely when $0 < \eta < 2/\lambda_{\max}(\tilde{H})$. In that range, its noise-driven second moment converges to the solution of*

$$S = (I - \eta \tilde{H}) S (I - \eta \tilde{H}) + \eta^2 \tilde{\Sigma}. \quad (34)$$

Let $Q^\top \tilde{H} Q = \text{diag}(h_i)$, $\nu_i = (Q^\top \tilde{\Sigma} Q)_{ii}$, and $b_i = \mathbb{E}[(Q^\top P^{-1/2} x_0)_i^2]$. For every integer $t \geq 0$ in the stable stepsize range,

$$\mathbb{E} f(x_t) = \frac{1}{2} \sum_i h_i (1 - \eta h_i)^{2t} b_i + \frac{\eta}{2} \sum_i \frac{\nu_i [1 - (1 - \eta h_i)^{2t}]}{2 - \eta h_i}. \quad (35)$$

The first term is the initial-state contribution; for deterministic x_0 it is the loss of the mean trajectory. The second term is the accumulated-noise contribution. Consequently,

$$\lim_{t \rightarrow \infty} \mathbb{E}f(x_t) = \frac{\eta}{2} \sum_i \frac{\nu_i}{2 - \eta h_i}. \quad (36)$$

Proof. Each deterministic eigenmode has multiplier $a_i = 1 - \eta h_i$. Strict stability is equivalent to $|a_i| < 1$. Conditional centering eliminates state-noise cross terms, so, for $y_t = Q^\top z_t$, $\mathbb{E}[y_{t,i}^2] = a_i^{2t} b_i + \eta^2 \nu_i \sum_{k=0}^{t-1} a_i^{2k}$. The geometric sum and $1 - a_i^2 = \eta h_i(2 - \eta h_i)$ give (35) after multiplication by $h_i/2$ and summation. The full second-moment recursion likewise has a convergent matrix series with unique limit (34). Since each b_i is finite, taking $t \rightarrow \infty$ proves (36). $\square$

The update-noise covariance is $\eta^2 P \Sigma P$ in the original coordinates and $\eta^2 \tilde{\Sigma}$ in z coordinates. Noise amplification depends on which covariance is measured.

Example 5.5 (Identical spectra, different stationary losses). Take $P = I$, $H = \text{diag}(1, 3)$, and $\eta = 1/2$. For $\Sigma_1 = \text{diag}(1, 0)$, (36) is $1/6$; for $\Sigma_2 = \text{diag}(0, 1)$, it is $1/2$. The covariance spectra are identical, as is the curvature spectrum, but their relative orientation changes the loss floor by a factor of three.

Adaptive P_t introduces changing coordinates, and current-sample dependence creates noise correlations. These effects and Muon’s nonlinear input map lie outside the fixed-preconditioner model, which also makes no claim about the generalization effects of reducing blockwise spectral disparity.

6 Comparing theoretical guarantees

6.1 Assumptions and convergence measures

Comparing convergence guarantees requires specifying the algorithm, objective class, geometry, oracle, schedule, and conclusion. We use $\Delta = F(W_1) - F_* > 0$, L_F for Frobenius smoothness, and L_{op} for spectral-to-nuclear smoothness. In particular,

$$\|\nabla F(X) - \nabla F(Y)\|_* \leq L_{\text{op}} \|X - Y\|_{\text{op}} \quad (37)$$

differs from Frobenius Lipschitz continuity. Translating between norms introduces dimension or rank factors; adaptive and non-Euclidean geometries require further distinctions [11].

Similarly, the statements

$$\frac{1}{T} \sum_t \mathbb{E} \|\nabla F(W_t)\|_F^2 \leq \varepsilon^2, \quad \frac{1}{T} \sum_t \mathbb{E} \|\nabla F(W_t)\|_* \leq \varepsilon, \quad \|\nabla F(W_t)\|_F \rightarrow 0 \text{ a.s.} \quad (38)$$

control different quantities. An average bound applies to a uniformly sampled iterate and implies neither last-iterate nor parameter convergence. Convex objective-gap and online regret results also have different interpretations. Table 3 summarizes the guarantees; appendix B records their assumptions and schedules.

Table 3: Representative proof map. The detailed assumptions and schedules in [appendix B](#) are essential to each row; rates are not a cross-method ranking.

Anchor	Controlled quantity	Main proof ingredient	Main coverage boundary
Original Shampoo [3]	Convex regret	Monotone metric potential and rank-sensitive domination	Cumulative statistics; exact powers
Structured / ASGO [18, 19]	Regret or convex gap	Restricted metric families; mode smoothness and noise traces	Selected one-sided/structured geometry; radius conditions
Two geometries [11]	Induced dual-gradient norm	Adaptive smoothness; structured trace bounds	Strong almost-sure noise envelope; selected metric cones
AdamW-style Shampoo v4 [54]	Nuclear gradient; generalized power measure	Matrix inequalities; momentum tracking	Coupled decay regime; fixed EMA for no-decay extension
Generalized SOAP [13]	Average nuclear gradient	Row-column trace control	Exact stated recursion, damping and schedule; no transport-quality guarantee
Taylor NS Muon [32]	Average nuclear gradient	Polar residual and momentum tracking	Particular Taylor family; spectral residual controls
Practical Muon [31]	Average Frobenius gradient	Actual polynomial and input tracking	Five empirical steps; horizon-dependent floor, momentum, decay
Layer-wise finite Muon [33]	Sum of average block Frobenius gradients	Joint descent, scalar-map alignment/energy, Nesterov tracking	Coupled schedules; exact arithmetic; constants fixed for dimension claim
MVR1 / MVR2 [34]	Average Frobenius gradient; PL gap with extras	Recursive estimator; finite-map descent	Changed oracle for MVR2; stronger assumptions for PL
Regularized proxy [52]	Deterministic gradient limit; PL linear gap	Bounded preconditioner and Lyapunov function	Soft proxy and admissible parameters, not equality to finite NS

6.2 The original Shampoo regret mechanism

Let f_t be convex losses, $G_t \in \partial f_t(W_t)$, and $\mathcal{R}_T = \sum_{t=1}^T [f_t(W_t) - f_t(W_*)]$. Suppose $r \geq 1$, $\text{rank}(G_t) \leq r$ and $\|W_t - W_*\|_F \leq D$. For the cumulative update (12), the original Shampoo analysis yields the following form [3, Theorem 7]:

Theorem 6.1 (Cumulative Shampoo regret, common notation). *With exact matrix powers, fixed stepsize $\eta > 0$, and the assumptions above,*

$$\mathcal{R}_T \leq \left(\frac{D^2}{2\eta} + \eta r \right) \text{tr}(L_T^{1/4}) \text{tr}(R_T^{1/4}). \quad (39)$$

Choosing $\eta = D/\sqrt{2r}$ gives

$$\mathcal{R}_T \leq \sqrt{2r} D \text{tr}(L_T^{1/4}) \text{tr}(R_T^{1/4}). \quad (40)$$

Metric projection onto a bounded convex domain can enforce the distance condition; without projection, it must hold along the iterates. The tuning in (40) requires a radius bound.

Proof mechanism. Set $g_t = \text{vec}(G_t)$, $H_t = R_t^{1/4} \otimes L_t^{1/4}$, and $S_t = \varepsilon I + (1/r) \sum_{s \leq t} g_s g_s^\top$. A rank-sensitive domination gives

$$S_t \preceq I_n \otimes L_t, \quad S_t \preceq R_t \otimes I_m, \quad S_t^{1/2} \preceq H_t. \quad (41)$$

The final step uses monotonicity of the positive matrix geometric mean and of the square root. A variable-metric potential controls the displacement term by $D^2 \text{tr}(H_T)/(2\eta)$, while trace concavity gives $\sum_t g_t^\top H_t^{-1} g_t \leq 2r \text{tr}(H_T)$. Combining them proves (39); [appendix A.1](#) supplies the intermediate inequalities.

Cumulative factors increase in Loewner order, whereas EMA factors may shrink, introducing positive metric variation into the potential. Extending [theorem 6.1](#) to grafting, momentum, stale powers, or rounding likewise requires additional bounds.

6.3 Structure-sensitive convex results and practical nonconvex Shampoo

One-sided structured adaptation. Xie et al. [19] analyze algebraically structured adaptive preconditioner families. Their one-sided regret bound is proportional to

$$D_{\text{op}} \text{tr} \left(\left[\sum_{t=1}^T G_t G_t^\top \right]^{1/2} \right), \quad (42)$$

plus initialization terms, under explicit factor normalization and domain-radius conditions. The matrix-subalgebra framework covers admissible families; membership must be established separately for two-sided separable operators.

For ASGO [18, Theorem 3], suppose F is convex and one-smooth in $\|X\|_A^2 = \text{tr}(X^\top A X)$ for SPD A , and a single-sample noise matrix obeys $\mathbb{E}[NN^\top] \preceq V^2$. With minibatch size B , its prescribed stepsize and damping, and bounded iterate radii D_{op} and D_F , the reported bound is

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[F(W_t) - F(W_*)] \leq \frac{4D_{\text{op}}^2 \text{tr}(A)}{T} + \frac{2\sqrt{2}D_{\text{op}} \text{tr}(V)}{\sqrt{BT}} + \frac{2\varepsilon D_F^2}{D_{\text{op}} T}. \quad (43)$$

Curvature and noise enter through traces in the selected mode geometry. Low effective spectral complexity can improve these quantities relative to isotropic bounds, subject to the stated geometry and radius assumptions.

Adaptive and non-Euclidean smoothness. Xie et al. [11, Theorems C.7–C.8] obtain $\tilde{\mathcal{O}}(T^{-1/4})$ stationarity bounds for specified adaptive families, including one-sided Shampoo, in nonconvex cumulative and EMA settings. Adaptive smoothness optimizes over admissible quadratic geometries, whereas normalized steepest descent uses a fixed norm. Assumption C.1 requires an almost-sure two-sided envelope on stochastic gradient outer products, stronger than a conditional covariance bound. Theorems 4.6 and 4.9 distinguish adaptive-noise upper bounds from dimension-dependent standard-noise lower bounds for sign descent. Two-sided factors and finite-NS maps require separate family and oracle arguments.

EMA, momentum, and decoupled decay. Li et al. [54] analyze the explicit family

$$\begin{aligned} M_t &= \theta M_{t-1} + (1 - \theta)G_t, \\ L_t &= \beta L_{t-1} + (1 - \beta)G_t G_t^\top, \\ R_t &= \beta R_{t-1} + (1 - \beta)G_t^\top G_t, \\ W_{t+1} &= (1 - \eta\lambda)W_t - \eta(L_t + \varepsilon I)^{-1/(2p)} M_t (R_t + \varepsilon I)^{-1/(2q)}. \end{aligned} \quad (44)$$

For $p, q \in [1, \infty]$ with $p^{-1} + q^{-1} = 1$, the symmetric choice $p = q = 2$ has fourth roots; an endpoint is one-sided. We use $1/\infty = 0$ and interpret a zero matrix exponent as the identity. Under Frobenius smoothness, conditional unbiasedness, matrix noise envelopes, a lower eigenvalue bound $\hat{\varepsilon}$, and the coupled parameters in [appendix B.4](#), Theorem 1 of the inspected v4 gives

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} \|\nabla F(W_t)\|_* \leq \left(8\sqrt{m+n} + \frac{119\hat{\sigma}}{\sqrt{\hat{\varepsilon}}} \right) \max \left\{ \left(\frac{\sigma^2 L_F \Delta}{T} \right)^{1/4}, \sqrt{\frac{L_F \Delta}{T\gamma}} \right\}. \quad (45)$$

Here $\gamma \in (0, 1)$ and $\hat{\sigma}^2 = \max\{\sigma^2, L_F \Delta / (T\gamma^2)\}$. The learning rate, momentum, damping, decay, and initialization are jointly constrained.

Theorem 2 of v4 permits $p^{-1} + q^{-1} = 2 - \omega$, $0 < \omega \leq 2$, and a fixed second-moment EMA coefficient $\beta \in (0, 1)$. It requires $\lambda = 0$ and horizon-dependent first-moment momentum, stepsize, and damping. It controls $\sum_i \sigma_i (\nabla F)^\omega$ at order $\mathcal{O}(T^{-\omega/4} + T^{-1/2})$ with problem constants fixed. Neither theorem covers arbitrary fixed schedules with decay, cached powers, or mixed precision.

6.4 Rotation-based convergence and information transport

Sun et al. [13, Algorithm 4; Theorem 1] analyze generalized SOAP:

$$\begin{aligned} M_t &= \theta M_{t-1} + (1 - \theta)G_t, \\ V_t &= \beta V_{t-1} + (1 - \beta)(Q_{L,t}^\top G_t Q_{R,t})^{\odot 2}, \\ W_{t+1} &= W_t - \eta Q_{L,t} \frac{Q_{L,t}^\top M_t Q_{R,t}}{\sqrt{V_t + \varepsilon}} Q_{R,t}^\top. \end{aligned} \quad (46)$$

First moments remain in original coordinates; orthogonal rotations may depend on G_t . With Frobenius smoothness, unbiased variance-bounded gradients and the dossier's schedule,

$$\frac{1}{T} \sum_t \mathbb{E} \|\nabla F(W_t)\|_* \leq \left(2\sqrt{m+n} + \frac{16\hat{\sigma}}{\delta} \right) \left(\frac{\hat{\sigma}^2 L_F \Delta}{T} \right)^{1/4}. \quad (47)$$

Row-column trace control proves convergence without reconstructing the historical second moment. Because the bound also holds for identity rotations, it establishes no advantage for learned bases.

The theorem retains coordinate-indexed V_t across rotations and uses the stated normalizer, without weight decay or bias correction. Transporting the stored state changes the recurrence. [Proposition 7.4](#) addresses recovery of the historical operator, a separate question from convergence.

6.5 Exact polar, convergent NS, and actual finite maps

An exact-gradient baseline. For deterministic $D_t = \text{polar}(\nabla F(W_t))$, L_F -smoothness gives

$$F(W_{t+1}) \leq F(W_t) - \eta \|\nabla F(W_t)\|_* + \frac{1}{2} L_F \eta^2 r. \quad (48)$$

A constant horizon-dependent stepsize yields an average nuclear gradient bound $\sqrt{2L_F r \Delta/T}$. Under (37), the remainder is $L_{\text{op}} \eta^2/2$, yielding $\sqrt{2L_{\text{op}} \Delta/T}$. The absence of r in the second bound comes with a different smoothness constant, L_{op} in place of L_F . Stochastic momentum also requires an input-tracking bound.

Convergent Taylor-family NS. Kim and Oh [32, Theorems 1–2] analyze a specified Taylor polynomial family, operator-to-nuclear smoothness, unbiased minibatch gradients, and horizon-dependent momentum. In our notation their stationarity estimate has the form

$$\frac{1}{T} \sum_t \mathbb{E} \|\nabla F(W_t)\|_* \leq \chi_K \mathcal{O} \left(\sqrt{\frac{L_{\text{op}} \Delta}{T}} + \frac{\sigma r}{\sqrt{BT}} + \left(\frac{r \sigma^2 L_{\text{op}} \Delta}{BT} \right)^{1/4} \right). \quad (49)$$

The inner-iteration factor χ_K depends on the initialized support residual and is uniformly bounded only when that residual is uniformly below one. Appendix H.4 excludes the empirical coefficient triple from the Taylor family, so (49) does not cover the empirical quintic.

Actual empirical polynomials. Wang et al. [31, Theorem 1] retain the five-step empirical polynomial, a Nesterov-style input, normalization floor, and decay. Their nonconvex average Frobenius-gradient rate is $\mathcal{O}(T^{-1/4})$ under schedules including $\eta \asymp T^{-3/4}$, $1 - \beta \asymp T^{-1/2}$, and a floor of order at most $T^{-1/4}$, with a prescribed coupled decay choice.

Peng et al. [33, Theorem 3.4 and Corollary 3.5] analyze layer-wise finite maps and Nesterov tracking under generalized smoothness, in exact arithmetic and without weight decay. They obtain a $T^{-1/4}$ average Frobenius bound with coupled horizon-dependent coefficients. The bound has no explicit matrix dimension or rank when block count, scalar-map constants, smoothness, and noise are fixed; these quantities may themselves vary with network width.

These analyses control the finite map’s alignment and energy through scalar inequalities, as in the following lemma.

Lemma 6.2 (Finite-map alignment and descent). *Let q be a scalar function with $q(0) = 0$ and $0 < \ell \leq q(s)/s \leq u$ for $0 < s \leq 1$. For $Z \neq 0$, define $\mathcal{T}(Z) = U \text{diag}(q(\sigma_i/\|Z\|_F))V^\top$ and $\mathcal{T}(0) = 0$. For any matrix g of the same shape,*

$$\begin{aligned} \|\mathcal{T}(Z)\|_F &\leq u, & \langle Z, \mathcal{T}(Z) \rangle &\geq \ell \|Z\|_F, \\ \langle g, \mathcal{T}(Z) \rangle &\geq \ell \|g\|_F - (\ell + u) \|Z - g\|_F. \end{aligned} \quad (50)$$

If $g = \nabla F(W)$ and F is L_F -smooth, then

$$F(W - \eta \mathcal{T}(Z)) \leq F(W) - \eta \ell \|g\|_F + \eta(\ell + u) \|Z - g\|_F + \frac{1}{2} L_F \eta^2 u^2. \quad (51)$$

Proof. For $s_i = \sigma_i/\|Z\|_F$, $\sum_i s_i^2 = 1$. Thus $\sum_i q(s_i)^2 \leq u^2$ and $\sum_i \sigma_i q(s_i) \geq \ell \|Z\|_F$. Subtract $\langle Z - g, \mathcal{T}(Z) \rangle$ and use $\|Z\|_F \geq \|g\|_F - \|Z - g\|_F$. The zero-input case obeys the same inequality. Smoothness proves the final line. $\square$

For exact gradients, averaging (51) and optimizing the constant stepsize yields $(u/\ell)\sqrt{2L_F\Delta/T}$. Momentum requires a tracking bound. Scalar-map bounds for the practical polynomial are available [33, 34]; a normalization floor requires a separate argument near zero.

6.6 Tracking, variance reduction, and asymptotic claims

The EMA recurrence exhibits the noise–drift trade-off. For $e_t = M_t - \nabla F(W_t)$,

$$e_t = \beta e_{t-1} + (1 - \beta)N_t + \beta(\nabla F(W_{t-1}) - \nabla F(W_t)). \quad (52)$$

Large β suppresses fresh noise but retains drift. With bounded updates, typical noise and drift terms scale as $\sqrt{1 - \beta}$ and $\eta/(1 - \beta)$, respectively. Coupling η and β controls both; a theorem sending β toward one does not justify an arbitrary fixed β as T grows.

Chang et al. [34, Theorems 6 and 8] obtain anytime $\tilde{\mathcal{O}}(T^{-1/4})$ for MVR1 and $\tilde{\mathcal{O}}(T^{-1/3})$ for MVR2 with practical finite NS and controlled decay. MVR2 requires a variance-reduced estimator, mean-square smoothness of same-sample gradient differences, and evaluation of each fresh sample at two parameter locations. Its faster rate therefore entails additional oracle work. Their PL-based last-iterate objective bounds also require bounded gradients along the iterates.

The Frobenius Polyak–Łojasiewicz (PL) condition is $\frac{1}{2} \|\nabla F(W)\|_F^2 \geq \mu_{\text{PL}}(F(W) - F_*)$ for some $\mu_{\text{PL}} > 0$ on the relevant domain. This additional assumption connects stationarity to objective error.

The soft polar proxy $(MM^\top + \varepsilon^2 I)^{-1/2}M$ admits a bounded-preconditioner heavy-ball interpretation and deterministic Lyapunov and PL arguments under specified parameter conditions [52]. These results concern a different map from the empirical finite polynomial. Counterexamples for specific Muon settings identify further limits [53]. For example, $x_{t+1} = x_t - \eta \text{sign}(x_t)$ on $f(x) = x^2/2$ cycles between x_0 and $x_0 - \eta$ when $0 < x_0 < \eta$; fixed normalized steps need not yield vanishing gradients.

7 Inexact matrix computation

7.1 Matrix targets and numerical routines

An eigendecomposition $A = Q\Lambda Q^\top$ gives the SPD matrix function $A^{-p} = Q\Lambda^{-p}Q^\top$; an SVD gives a rectangular matrix's canonical polar factor. QR produces an orthogonal basis and triangular factor and can track eigenvectors through orthogonal iteration. Newton–Schulz denotes a family of polynomial or rational recurrences. Specifying the routine alone therefore leaves the target operator ambiguous.

Proposition 7.1 (A QR basis is not generally a polar factor). *If $Z = QR$ is a thin QR factorization with $Q^\top Q = I$ and R invertible, then $\text{polar}(Z) = Q \text{polar}(R)$. In particular, for*

$$Z = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad Q = I, \quad \text{polar}(Z) = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix} \neq Q. \quad (53)$$

Proof. Since $Z^\top Z = R^\top R$, $Z(Z^\top Z)^{-1/2} = QR(R^\top R)^{-1/2}$. For the displayed matrix, the proposed polar factor is orthogonal and its transpose times Z is SPD, which identifies the polar decomposition. $\square$

SOAP- and KL-based implementations use QR to update a historical basis [5, 10]. A step $Q^+ = \text{qr}(AQ)$ approximates an eigenbasis of A with accuracy depending on the previous basis and spectrum. Caching that basis while refreshing scales differs from caching the inverse-root operator.

Table 4: Numerical targets and the approximation that must be controlled.

Target	Representative realization	Main error object	Dense cost scale
SPD inverse power	EVD or inverse-root solver	$\ \widehat{A^{-p}} - A^{-p}\ $	$O(d^3)$ for dense EVD
Moving eigenbasis	EVD refresh; warm-started QR iteration	Operator error and basis/subspace mismatch	$O(d^3)$ for a full QR step; less for restricted subspaces
Rectangular polar	SVD; convergent NS; Polar Express	Polar residual or singular response	$O(mn \min(m, n))$ per dense SVD-scale operation or polynomial step
Practical finite map	Prescribed polynomial with finite K	Alignment, energy, rounding of that map	$O(Kmn \min(m, n))$

7.2 Fixed-depth polar approximation

Proposition 7.2 (A boundary of fixed-polynomial approximation). *Let q be any fixed continuous scalar map with $q(0) = 0$. Normalize $Z_\varepsilon = \text{diag}(1, \varepsilon)$ by its Frobenius norm and apply q to its singular values. Then*

$$\liminf_{\varepsilon \downarrow 0} \|\mathcal{T}(Z_\varepsilon) - \text{polar}(Z_\varepsilon)\|_{\text{op}} \geq 1. \quad (54)$$

Proof. The second normalized singular value is $\varepsilon/\sqrt{1+\varepsilon^2} \rightarrow 0$, so its transformed value tends to zero. For every $\varepsilon > 0$, the corresponding exact polar singular value is one. The spectral error is at least their absolute difference. $\square$

Any fixed finite polynomial composition has this limitation: arbitrarily small positive singular values defeat uniform polar approximation at fixed depth. Convergence as depth increases at a fixed input remains possible. Uniform guarantees require spectral bounds, adaptive work, or a different error measure.

Scalar-map inequalities instead yield descent through [lemma 6.2](#), as in direct analyses of empirical Muon polynomials [\[31, 33, 34\]](#). Rank-independent bounds can have large constants because bounds on $q(s)/s$ may grow with depth. If $|q(s)| \leq c$ as well, the update norm is bounded by $\min\{u, c\sqrt{r}\}$; the tighter estimate depends on depth and matrix shape.

For example, let $q_5 = \phi^{\circ 5}$ for the empirical quintic $\phi(s) = 3.4445s - 4.7750s^3 + 2.0315s^5$. Direct evaluation and the chain rule give

$$q'_5(0) = 3.4445^5 \approx 484.8763, \quad q_5(1) \approx 0.6964364, \quad \frac{u}{\ell} \geq \frac{q'_5(0)}{q_5(1)} \approx 696.225. \quad (55)$$

Any constants satisfying [lemma 6.2](#) obey $u \geq \lim_{s \downarrow 0} q_5(s)/s = q'_5(0)$ and $\ell \leq q_5(1)$. The resulting lower bound concerns the lemma's uniform constants for these coefficients and depth, rather than observed optimization error.

Improving the inner initialization. Turbo-Muon preconditions the Newton–Schulz initialization to reduce inner work [\[55\]](#). Comparing such schemes requires specifying the initialized matrix, finite map, and outer-loop progress, since fewer inner iterations can change the update and need not reduce total training time.

7.3 Damping and spectral sensitivity

Proposition 7.3 (Damped inverse-power perturbation). *For $A, B \succeq 0$, $\delta > 0$, and $0 < p \leq 1$,*

$$\|(A + \delta I)^{-p} - (B + \delta I)^{-p}\|_{\text{op}} \leq p\delta^{-p-1} \|A - B\|_{\text{op}}. \quad (56)$$

The resolvent proof in [appendix A.3](#) handles noncommuting A and B . For $p = 1$, the inverse identity suffices.

Clustered eigenvalues allow large eigenvector rotations while smooth functions of the full matrix remain stable. Small damping, however, magnifies operator sensitivity even with an accurate eigendecomposition. EVD error, eigenbasis tracking error, and inverse-root update error measure different effects.

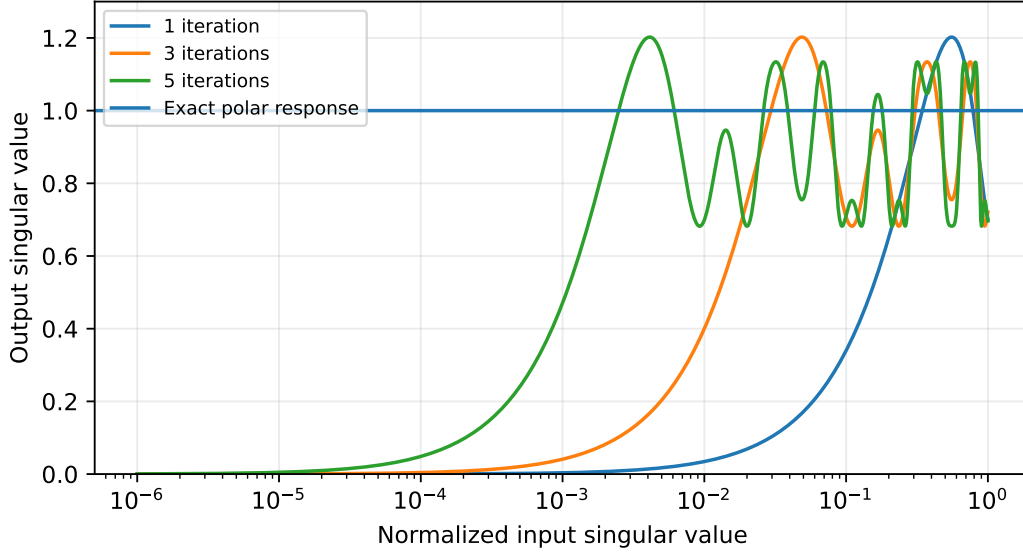


Figure 1: Analytic singular responses of the empirical Muon quintic after one, three, and five steps. One is not a fixed point of this map, and tiny input singular values remain suppressed at fixed depth.

For $D = AZB$ and $\hat{D} = \hat{A}Z\hat{B}$,

$$\|\hat{D} - D\|_F \leq \|\hat{A} - A\|_{\text{op}} \|Z\|_F \|B\|_{\text{op}} + \|\hat{A}\|_{\text{op}} \|Z\|_F \|\hat{B} - B\|_{\text{op}}. \quad (57)$$

Combining (56) and (57) bounds update error from Gram-estimation error, with constants depending on damping. Purifying Shampoo, FOAM, low-precision reparametrization, and DASH address errors in estimation, stale state, or numerical execution [21, 27–29].

7.4 Information loss under basis changes

Proposition 7.4 (Diagonal state cannot determine arbitrary rotated diagonals). *There is no rule that, given only $\text{diag}(C)$ and an arbitrary orthogonal Q , recovers $\text{diag}(Q^\top C Q)$ for every positive-definite C .*

Proof. For $|\rho| < 1$, let

$$C_\rho = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \quad Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \quad (58)$$

All inputs have diagonal $(1, 1)$, but $\text{diag}(Q^\top C_\rho Q) = (1 + \rho, 1 - \rho)$. The proposed rule receives identical data for distinct required outputs. $\square$

Corollary 7.5 (Exact diagonal transport for all moments). *For a fixed orthogonal $Q \in \mathbb{R}^{d \times d}$, the vector $\text{diag}(Q^\top C Q)$ is determined by $\text{diag}(C)$ for every SPD C if and only if Q is a signed permutation matrix.*

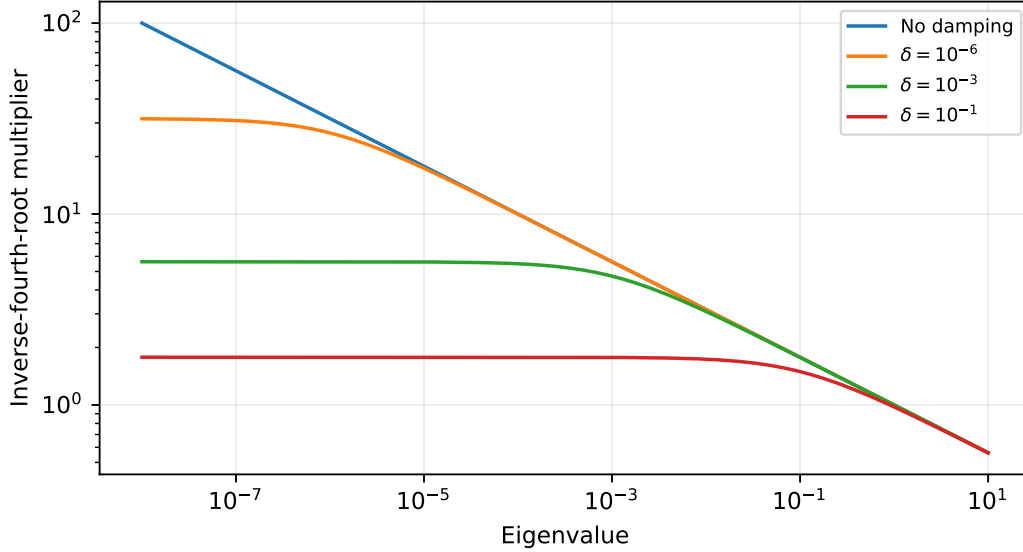


Figure 2: Analytic inverse-fourth-root multipliers at different damping values. Damping changes the target operator near small eigenvalues.

Proof. Signed permutations only permute diagonal entries. Conversely, for distinct indices a, b , the SPD matrices $I \pm \epsilon(e_a e_b^\top + e_b e_a^\top)$, $0 < \epsilon < 1$, have the same diagonal. Equality of their required outputs forces $Q_{aj}Q_{bj} = 0$ for every column j . Thus each column has at most one nonzero entry. Unit column norms and orthogonality force these entries to be ± 1 in distinct rows, proving the claim. $\square$

Diagonal storage discards cross-coordinate second moments. For $v = \text{diag}(C)$, a common transport model is

$$\hat{v}^+ = (Q \odot Q)^\top v, \quad v^+ - \hat{v}^+ = \text{diag}\left(Q^\top [C - \text{diag}(v)] Q\right), \quad (59)$$

where $\odot$ denotes entrywise multiplication and $v^+ = \text{diag}(Q^\top C Q)$. The rule is exact when the old full moment is diagonal, a condition that diagonal storage alone cannot verify. Preserving the historical operator therefore requires accounting for the omitted term or storing additional information. Sign and permutation alignment prevents reordering errors but cannot recover omitted moments under genuine rotations, even within repeated-eigenvalue subspaces. The index-retained recurrence in [section 6.4](#) admits a convergence proof without reconstructing that operator.

7.5 From numerical error to optimization progress

For an ideal update D and its realization $\hat{D} = D + E$, Frobenius smoothness gives

$$\begin{aligned} F(W - \eta \hat{D}) &\leq F(W) - \eta \langle \nabla F(W), D \rangle + \eta \|\nabla F(W)\|_F \|E\|_F \\ &\quad + \frac{L_F \eta^2}{2} (\|D\|_F + \|E\|_F)^2. \end{aligned} \quad (60)$$

Error reduces alignment and increases update energy. If $\langle \nabla F, D \rangle \geq a \|\nabla F\|_F$ and $\|E\|_F \leq e < a$, the linear descent coefficient remains at least $a - e$. Error bounds must therefore be assessed relative to the descent margin.

An error decomposition can telescope a population target, structured fit, online estimate, stale representation, finite map, and distributed realization. The triangle inequality bounds their sum without assuming independence; individual bounds may use different metrics. When proximity to exact polar is unavailable, direct alignment analysis is preferable to (60). For the unregularized objective, decay belongs in the designated update as $D + \lambda W$.

Roundoff can break symmetry or positive definiteness of accumulated statistics. Symmetrization, clipping, and extra damping alter the realized operator. Numerical analyses such as Polar Express and solver-focused systems work [29, 35] complement outer-loop theorems. Their error tolerances should reflect the update's descent margin and energy.

8 Distributed realizations and structural fidelity

Distributed Shampoo, scalable Muon, NorMuon, and Dion support large-scale matrix-structured optimization [8, 26, 41, 45]. Distributed computation may preserve the logical update or change its statistical input, operator, or state. *Structural fidelity* requires the update and its information to remain identifiable after sharding and communication are specified.

8.1 Aggregation order changes the statistic

Proposition 8.1 (Local second moments differ from the global-mean second moment). *For matrices G_j and nonnegative weights w_j summing to one, set $\bar{G} = \sum_j w_j G_j$. Then*

$$\sum_j w_j G_j G_j^\top - \bar{G} \bar{G}^\top = \sum_j w_j (G_j - \bar{G})(G_j - \bar{G})^\top \succeq 0. \quad (61)$$

Proof. Expand the right side and cancel the two cross terms using $\sum_j w_j G_j = \bar{G}$. $\square$

Averaging local Gram matrices adds a between-worker variation term to the Gram matrix of the averaged gradient. These statistics estimate different objects, with potentially large differences on heterogeneous data. The same holds on the right mode.

Nonlinear update maps also fail to commute with averaging. For $G_1 = \text{diag}(2, 1)$ and $G_2 = \text{diag}(-1, 1)$,

$$\text{polar}\left(\frac{G_1 + G_2}{2}\right) = I, \quad \frac{\text{polar}(G_1) + \text{polar}(G_2)}{2} = \text{diag}(0, 1). \quad (62)$$

Averaging locally orthogonalized momentum and orthogonalizing global momentum therefore define different updates.

8.2 An exact row-sharded polynomial construction

Some nonlinear global maps can be computed through distributed sufficient products without gathering the matrix onto one device.

Proposition 8.2 (Exact row-sharded polynomial iteration). *Partition X by rows as $X = (X_1^\top, \dots, X_p^\top)^\top$. Define $H = X^\top X = \sum_j X_j^\top X_j$. For any fixed polynomial π , the block updates $X_j^+ = X_j \pi(H)$ jointly equal $X^+ = X \pi(X^\top X)$. Repeating this construction, with a newly aggregated Gram matrix at every inner step and identical scalar coefficients, implements the corresponding global polynomial iteration exactly in real arithmetic.*

Proof. The identity for H follows by block multiplication. Stacking the local outputs gives $X \pi(H)$. Applying the same argument inductively proves equality at each subsequent inner step. A global normalization is also exact if its norm is computed from the sum of local squared Frobenius norms. $\square$

The left-written quintic in (23) equals $X[aI + b(X^\top X) + c(X^\top X)^2]$, so the proposition applies. Its efficiency relative to an all-gather depends on shape, shard orientation, and communication.

For an $m \times n$ row-sharded matrix on p workers, the construction replicates an $n \times n$ Gram matrix, requiring $O(mn/p + n^2)$ update workspace per worker. Each inner step uses $O(mn^2/p)$ dense arithmetic for local Grams and right products; $X_j H^2$ can be evaluated as $(X_j H) H$ without forming H^2 . Each step requires one Gram all-reduce, with an initial scalar reduction for normalization. An uncompressed ring all-reduce moves approximately $2(p-1)n^2/p$ words per worker per Gram, under the usual ring schedule. With per-message latency α and per-word cost β_{net} , that schedule has the simple model

$$2(p-1)\alpha + \frac{2(p-1)}{p}n^2\beta_{\text{net}} \quad (63)$$

per Gram collective, ignoring overlap and topology effects. This execution model is neither a communication lower bound nor a hardware benchmark. All-gather communicates on the scale of the matrix itself; neither scheme dominates across all shapes and parallel layouts.

8.3 Storage and geometry partitioning

Sharding storage preserves a logical matrix’s geometry; independent block transforms change it. Replacing a global left factor by its block diagonal, for example, removes cross-block correlations. A one-sided rectangular transform, blocked two-sided Shampoo, and a full transform with sharded state are distinct constructions.

For historical Shampoo factors, a right Gram of a row-sharded gradient can be obtained by summation, whereas the full left Gram contains off-diagonal row-block products $G_i G_j^\top$. Local $G_i G_i^\top$ alone cannot represent the full left factor exactly. Assigning complete layer factors to different workers distributes state without changing each layer’s operator. Implementations must specify their partition [26].

Low-rank updates can restrict the transformed subspace and carry residual state over time, as in Dion [45, 46]. Bounding error in the accumulated parameter update requires tracking omitted components and subspace evolution, beyond the communicated rank. The same applies to subspace basis updates for Shampoo/SOAP [28].

8.4 Practical layouts and communication-reducing algorithms

Data parallelism (DP) aggregates replicated gradient contributions; tensor parallelism (TP) splits a logical matrix along model dimensions; ZeRO-3 partitions parameters, gradients, and optimizer state, potentially across flattened buffers. A storage shard may lack the complete matrix required by an optimizer. Reconstruction, communication of sufficient products, and local transforms have different mathematical consequences.

Canzona decouples optimizer ownership from parameter sharding and pipelines TP reconstruction with load-balanced scheduling [56]. DMuon combines owner-based execution, shape-aware kernels, cost-based scheduling, and asynchronous parameter publication [57]. Both seek to preserve the update while changing execution. Owners must accommodate complete matrices, state, and transform workspace; unequal shapes create unequal costs, so balancing parameter counts need not balance peak memory or optimizer work. Asynchronous execution can overlap communication and computation while respecting dependencies, without making gradients, momentum, or parameters stale.

MatrixFSDP makes whole-matrix ownership a ZeRO-3 placement, so backward reduction delivers each complete gradient to its owner [58]. Its communication-free claim covers optimizer-step matrix collectives; forward/backward materialization and communication remain. The current owner planner excludes TP fragments and thus arbitrary combined TP/ZeRO-3 layouts.

HyperParallel-FSDP uses layout-derived communication groups, deduplicated orthogonalization, and shape-fused Newton–Schulz within a topology-aware runtime [59]. The reported results apply to the studied hardware and layouts. Placement and scheduling must account for fragmented buffers, owner memory, communication overlap, and ownership-aware checkpointing.

MuonBP alternates shard-local and full-matrix orthogonalization, with distinct step-size pre-descriptions [60]. EF21-Muon introduces bidirectional compression and error-feedback state into a non-Euclidean linear-minimization-oracle method [61]. These changes to the global Muon recurrence and estimator require separate analysis.

Placing sign compression before or after the spectral oracle can change whether descent is preserved, and output-side error feedback does not always repair lost descent [62]. Results for an exact spectral oracle also need separate treatment from finite-polynomial implementations.

8.5 Assumptions for distributed guarantees

A distributed theorem must specify parameter blocks and aliases, aggregation order, and ownership of historical state, including compression residuals. Collective products, synchronization and publication dependencies, parameter and gradient versions, and arithmetic precision then determine whether execution introduces rounding differences or changes the recurrence.

For an outer-loop proof assuming exact global matrix operations, [proposition 8.2](#) supplies an algebraic implementation argument. Blocking, compression, delayed communication, or local normalization that changes the operator requires an additional approximation or descent bound. Time-to-target comparisons can then distinguish faster execution of the same update from a better cost–progress trade-off for a changed update.

9 Published empirical evidence and its interpretation

Table 10 in appendix D records the source versions, experimental conditions, and locators for the six published observations considered here.

9.1 Separate timescales for bases and scales

On C4 with a 660M decoder-only model, SOAP and Shampoo are close at refresh interval 1. Both degrade as the interval increases to 100, with SOAP degrading more slowly. The study uses learning-rate and paired hyperparameter sweeps, mixed-bf16 training, fp32 factor updates, and QR-based basis refreshes after initialization [5, v2, Fig. 1 and Secs. 5–6]. This recipe allows less frequent basis refreshes while coordinatewise scales continue to adapt.

On Imagewoof, Purifying Shampoo evaluates SimpleViT and ConvNeXt V2 trained for 100 epochs. EShampoo matches or improves on Adam-grafted Shampoo in final training cross-entropy, refreshing roots or bases every 100 steps and correcting eigenvalues every step. Ungrafted Shampoo matches the ViT result but not the ConvNeXt V2 result [21, v1, Fig. 2 and App. E.1]. Separating basis refresh from scale correction is supported here; removing scale control remains architecture-dependent.

These observations motivate separate bounds for eigenbasis error, scale-estimation error, and state policy at a basis change. They do not isolate every component of historical-state transport: proposition 7.4 addresses its information limit, while section 6.4 gives a convergence statement.

9.2 Update scaling

Scalable Muon evaluates an approximately 800M dense model at 4B tokens of a 20B-token schedule. Validation loss is 2.812 with baseline scaling, 2.789 with RMS normalization, and 2.789 with a shape-adjusted learning rate. The recipe uses five Newton–Schulz steps and momentum 0.95; the panel-specific tuning budget, data, and precision were not extracted [8, v1, Table 1 and Sec. 3.1].

This ablation within one training regime supports retaining the block scale in (19): the polar direction alone cannot explain the recipe’s performance. Learning rates also need to be interpreted alongside update magnitudes.

9.3 Polar accuracy and training loss

Polar Express varies inner depth from 2 to 30 and tests exact SVD with GPT-2-Small on 1B FineWeb tokens. The ablation fixes learning rate 0.05 and weight decay 0.1, using four H100s, batch 32, mixed-bf16 training, and bf16 polynomial matrix-sign computation. Five or six polynomial steps improve final validation loss over two or three; further accuracy, including SVD, adds no loss improvement. SVD roughly doubles training-step runtime, including forward and backward computation [35, v5, Fig. 5 and Secs. 4.2–4.3].

The recipe-specific threshold is consistent with an analysis of finite-map alignment and energy (lemma 6.2 and proposition 7.2), but gives neither a universal iteration budget nor a general conclusion about exact polar updates.

9.4 Iteration and wall-clock efficiency

With 450M Llama3, KL-Shampoo reports better progress than Muon per iteration but worse progress per wall-clock time; both improve on SOAP. The panel uses 120 random-search runs per method and fp32 QR every ten iterations, with refresh frequency not tuned for runtime [10, v10, Fig. 9]. These results concern that schedule, distinct from the smaller-model, bf16 Fig. 1 experiments with 150 runs per method.

In DASH’s matched Llama-953M/C4 comparison, DIST and DASH both report mean validation perplexity 11.87, while optimizer-step latency falls from 675 to 221 ms. The row uses block size 2,048, fp32 coupled-Newton inverse roots, refresh every step, and Frobenius normalization; the study uses three seeds and Adam grafting with a transferred learning rate [29, v1, Table 1 and Sec. 5]. This latency excludes forward and backward passes, unlike the Polar Express runtime.

KL-Shampoo trades iteration efficiency against execution cost; DASH reduces optimizer time at the same reported quality. These timing results cannot be pooled across studies with different settings and timing denominators (appendix D).

10 Open problems and predictive theory

Predictive theory must identify when statistical information, geometry, and execution improve quality within a specified budget.

10.1 Statistical information and finite-time progress

In the frozen model (proposition 5.4), a structured transformation changes both initial-state contraction and orientation-weighted noise accumulation in (35). Minimizing the condition number alone does not optimize both terms. Minimizing the stationary noise floor alone can favor an arbitrarily small effective step despite slow finite-time progress.

One target is to minimize a finite-time loss bound over a class such as $P = B \otimes A$, with fixed normalization and a state or communication budget. Normalization removes the redundancy between the scalar scale of P and the learning rate. The result should identify conditions on curvature, noise orientation, and initialization under which separable or one-sided transformations improve the bound. History-dependent estimates further require control of their feedback with the parameter trajectory.

10.2 Basis memory and scale memory

Corollary 7.5 shows that, with diagonal state alone, signed permutations are precisely the orthogonal changes admitting exact transport for every SPD moment. Open questions concern restricted moment families, approximate transport under controlled rotation, and representations of omitted off-diagonal information. The missing term in (59) suggests a memory–error trade-off for diagonal-plus-low-rank states.

Generalized SOAP has a convergence theorem for moving, even current-sample-dependent, orthogonal bases under a specified recurrence and schedule [13]. Its identity-basis specialization

satisfies the same order of bound, leaving open which basis and state policy improves constants or transient progress. An analysis should separate operator preservation from stationarity and handle rotations within repeated eigenspaces through projectors or full operators where appropriate. Policies motivated by the base update, such as ARO, extend this problem beyond eigenbasis tracking [12].

10.3 Inner accuracy and error allocation

Fixed-depth polar approximation is nonuniform near vanishing singular values, whereas damped inverse powers have a damping-dependent operator Lipschitz bound (propositions 7.2 and 7.3). Accuracy criteria should preserve alignment and control update energy for the outer method. As (55) illustrates, even an explicit dimension-free bound may be too conservative to predict useful iteration budgets.

A stopping principle could compare numerical, estimation, and temporal errors in compatible update norms and allocate matrix products and synchronizations to the dominant term. Its guarantee should relate progress to gradient evaluations, matrix products, and communication, including low-precision positivity repairs and damping because they change the operator.

10.4 Guarantees for practical schedules

Finite-NS results retain empirical maps, generalized SOAP permits broad rotation choices, and AdamW-style Shampoo v4 covers fixed second-moment EMA coefficients in a no-decay general-exponent regime [54]. Other quantities still require coupled choices. Fixed second-moment memory, fixed first-moment momentum, positive decay, and a long constant-learning-rate phase remain separate questions.

Phase-wise guarantees for warmup, sustained learning, and cooldown should specify whether each phase approaches a noise-dependent neighborhood or yields a vanishing stationarity measure. For stale or grafted Shampoo, predictable-state estimates can be combined with positive metric variation ((69)) without assuming cumulative monotonicity. Moving-basis guarantees must identify the state recurrence controlled. Last-iterate and parameter convergence require arguments beyond average stationarity.

10.5 Distributed geometry and communication cost

Proposition 8.2 computes a global polynomial exactly through aggregated Gram products. Its benefit depends on shape, shard orientation, collective latency, and replicated Gram size. Jointly choosing the operator family and communication pattern requires identifying which cross-block interactions deliver enough progress to justify their cost.

Whole-matrix ownership and topology-aware layouts can reduce reconstruction and synchronization costs while preserving the designated update [58, 59]. Choosing ownership, shard orientation, and overlap under peak-memory and dependency constraints remains an open problem, as does predicting time to target. Optimizer-step communication savings must be assessed alongside training communication.

For compressed updates, bounding the accumulated effect of omitted directions or preserving a specified descent margin could support comparisons with exact, more expensive transforms. The state and arithmetic models in [table 7](#) and the ring schedule in [section 8](#) make these trade-offs explicit.

10.6 Conditions for predictive comparisons

The conditional conclusions in [table 5](#) offer analytical tests for specific regimes.

Table 5: Conditional conclusions for structured optimization.

Condition or diagnostic	Conclusion available	Quantity still needed
Same instantaneous input, no damping or history	Two-sided fourth roots equal canonical polar	Historical state and momentum for trajectory comparison
One mode admits favorable curvature/noise traces and bounded radii	ASGO’s bound isolates that mode’s structural complexity	Actual radii, traces, and arithmetic cost
Fixed finite spectral map with positive scalar alignment bounds	Descent can be analyzed without uniform polar approximation	Tracking error and usable scalar constants
Slowly changing basis with rapidly changing scales	Separate refresh and scaling policies are meaningful design choices	State policy and measured or bounded update error
Row-sharded input with affordable replicated Gram	A global polynomial can be computed exactly in real arithmetic	Collective cost and rounding error

Predictive theory should combine these tests with measured spectra, noise orientation, matrix shape, and resource budgets. Population risk also depends on data and parameterization. Connecting matrix geometry to implicit bias therefore requires these additional assumptions, beyond those supporting training-loss or stationarity results.

11 Conclusion

Shampoo, SOAP, and Muon exploit matrix structure through different statistics and update geometries. Algebraic equivalences depend on inputs, history, damping, and the numerical map. Instantaneous preconditioning and polar updates can coincide for matched inputs, while their historical variants need not follow the same trajectory. Finite spectral maps can support descent without uniformly approximating polar, and a factorized basis can support nonseparable adaptive scaling.

Whether these choices improve training enough to justify their cost remains unresolved. The frozen quadratic model shows how curvature and noise orientation affect finite-time loss. Changing bases, momentum, and estimated factors require analysis of their interaction with the parameter trajectory. Comparisons with AdamW should report time to a specified quality together with memory use, tuning budget, precision, and what the timing includes.

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A Proof details and auxiliary identities

A.1 The cumulative Shampoo potential

For the matrix case of Gupta et al. [3, Theorem 7], in the notation of theorem 6.1, all matrices are positive definite after initialization. Write $x_t = \text{vec}(W_t)$, $u = \text{vec}(W_*)$, and $g_t = \text{vec}(G_t)$. The update $x_{t+1} = x_t - \eta H_t^{-1} g_t$, with $H_t = R_t^{1/4} \otimes L_t^{1/4}$, satisfies the following inequality, also valid with H_t -metric projection onto a convex set containing u :

$$\langle g_t, x_t - u \rangle \leq \frac{\|x_t - u\|_{H_t}^2 - \|x_{t+1} - u\|_{H_t}^2}{2\eta} + \frac{\eta}{2} g_t^\top H_t^{-1} g_t. \quad (64)$$

Here $\|v\|_H^2 = v^\top H v$. Summing the numerator leaves initial and negative terminal terms and $\sum_{t=2}^T \|x_t - u\|_{H_t - H_{t-1}}^2$. Cumulative factors, their $1/4$ powers, and the Kronecker product increase in Loewner order, so $H_t \succeq H_{t-1}$. If $\|x_t - u\|_2 \leq D$, the displacement contribution is at most $D^2 \text{tr}(H_T)/(2\eta)$.

The rank-sensitive bound is

$$\frac{1}{r} g g^\top \preceq I_n \otimes G G^\top, \quad \frac{1}{r} g g^\top \preceq G^\top G \otimes I_m, \quad r \geq \text{rank}(G). \quad (65)$$

To prove the first inequality, g lies in the range of $B = I_n \otimes G G^\top$ and $g^\top B^\dagger g = \text{tr}[G^\top (G G^\top)^\dagger G] = \text{rank}(G)$. Applying Cauchy–Schwarz to $\langle B^{1/2} v, B^\dagger g \rangle$ proves $\langle v, g \rangle^2 \leq r v^\top B v$. The other inequality follows by transposition. Support pseudoinverses establish this inequality; the algorithmic preconditioners remain damped and invertible.

Let $S_t = \varepsilon I + (1/r) \sum_{s \leq t} g_s g_s^\top$, $A_t = I_n \otimes L_t$, and $B_t = R_t \otimes I_m$. Equation (65) gives $S_t \preceq A_t, B_t$. For SPD matrices, the geometric mean is $A \# B = A^{1/2} (A^{-1/2} B A^{-1/2})^{1/2} A^{1/2}$. Its monotonicity yields $S_t = S_t \# S_t \preceq A_t \# B_t$. The two Kronecker factors commute, so $A_t \# B_t = R_t^{1/2} \otimes L_t^{1/2} = H_t^2$. Square-root monotonicity and inverse order now give $S_t^{1/2} \preceq H_t$ and $H_t^{-1} \preceq S_t^{-1/2}$.

The scalar-valued function $X \mapsto \text{tr}(X^{1/2})$ is concave on SPD matrices, with differential $\frac{1}{2} \text{tr}(X^{-1/2} dX)$. Its supporting-hyperplane inequality at S_t implies

$$\text{tr}(S_t^{1/2}) - \text{tr}(S_{t-1}^{1/2}) \geq \frac{1}{2} \text{tr}[S_t^{-1/2} (S_t - S_{t-1})]. \quad (66)$$

Therefore,

$$\sum_{t=1}^T g_t^\top H_t^{-1} g_t \leq r \sum_{t=1}^T \text{tr}[S_t^{-1/2} (S_t - S_{t-1})] \quad (67)$$

$$\leq 2r \text{tr}(S_T^{1/2} - S_0^{1/2}) \leq 2r \text{tr}(H_T). \quad (68)$$

Convexity gives $f_t(W_t) - f_t(W_*) \leq \langle G_t, W_t - W_* \rangle$. Combining this with (64) and $\text{tr}(H_T) = \text{tr}(R_T^{1/4}) \text{tr}(L_T^{1/4})$ proves the stated regret bound and its optimized stepsize.

Nonmonotone metrics. For an arbitrary SPD sequence, the displacement contribution is bounded by

$$\frac{D^2}{2\eta} \left[\text{tr}(H_1) + \sum_{t=2}^T \text{tr}((H_t - H_{t-1})_+) \right], \quad (69)$$

where A_+ denotes the positive spectral part of a symmetric matrix. This follows from $v^\top A v \leq D^2 \operatorname{tr}(A_+)$ for $\|v\|_2 \leq D$. For EMA, basis changes, and approximate factors, (69) requires positive metric variation beyond endpoint control.

A.2 Separable moments and the population KL objective

Let $C = \mathbb{E}[\operatorname{vec}(G) \operatorname{vec}(G)^\top] \succ 0$ be finite, as in (17). Interpreting this possibly uncentered moment as an auxiliary Gaussian covariance imposes no zero-mean assumption on G . For SPD $A \in \mathbb{R}^{m \times m}$ and $B \in \mathbb{R}^{n \times n}$, the Gaussian KL objective depends on $B \otimes A$ through, up to a factor $1/2$,

$$J(A, B) = n \log \det A + m \log \det B + \mathbb{E} \operatorname{tr}(A^{-1} G B^{-1} G^\top). \quad (70)$$

The trace identity follows directly from column-major vectorization. Setting the A differential to zero gives $nA^{-1} - A^{-1} \mathbb{E}[G B^{-1} G^\top] A^{-1} = 0$, and likewise for B . Hence any interior stationary point satisfies

$$A = \frac{1}{n} \mathbb{E}[G B^{-1} G^\top], \quad B = \frac{1}{m} \mathbb{E}[G^\top A^{-1} G]. \quad (71)$$

The transformation $(A, B) \mapsto (cA, B/c)$ preserves the covariance and objective; normalization fixes this gauge. The equations characterize population stationarity, without establishing online convergence. For singular C , the expression remains a Gaussian cross-entropy fitting objective up to constants and scale. The KL divergence is infinite because the singular Gaussian occupies a proper subspace of zero probability under the SPD Gaussian, so discarding a finite constant cannot yield this objective. An interior minimizer may require regularization, and damping changes the objective or fixed-point map; see Lin et al. [10].

When $C = B \otimes A$ exactly, its (j, k) block is $B_{jk}A$. Summing the diagonal blocks proves $\mathbb{E}[G G^\top] = \operatorname{tr}(B)A$; the transposed contraction proves $\mathbb{E}[G^\top G] = \operatorname{tr}(A)B$. This proves the covariance-scale correction in (14) for exact separability; arbitrary mode-wise Gram matrices need not factorize the covariance.

A.3 Resolvent proof for damped inverse powers

For proposition 7.3, A and B need not commute. When $0 < p < 1$, spectral calculus gives

$$(A + \delta I)^{-p} = \frac{\sin(\pi p)}{\pi} \int_0^\infty t^{-p} (A + (\delta + t)I)^{-1} dt. \quad (72)$$

For $s = \delta + t > 0$, the resolvent identity is

$$(A + sI)^{-1} - (B + sI)^{-1} = (A + sI)^{-1} (B - A) (B + sI)^{-1}. \quad (73)$$

Both inverse factors have operator norm at most s^{-1} . The integral converges in operator norm, and hence

$$\|(A + \delta I)^{-p} - (B + \delta I)^{-p}\|_{\text{op}} \leq \frac{\sin(\pi p)}{\pi} \|A - B\|_{\text{op}} \int_0^\infty \frac{t^{-p}}{(\delta + t)^2} dt \quad (74)$$

$$= p \delta^{-p-1} \|A - B\|_{\text{op}}. \quad (75)$$

The last integral is $\delta^{-p-1} \Gamma(1-p) \Gamma(1+p)$, and the reflection identity gives the constant. For $p = 1$, the resolvent identity at $s = \delta$ proves the result directly.

A.4 A factorized basis need not define a separable operator

Let $Q = Q_R \otimes Q_L$ and $P = Q \operatorname{diag}(\operatorname{vec}(S))Q^\top$, with all $S_{ij} > 0$. In this fixed basis, P is a single SPD Kronecker product if and only if $S = ab^\top$ for positive vectors a, b . Sufficiency is immediate from $\operatorname{diag}(\operatorname{vec}(ab^\top)) = \operatorname{diag}(b) \otimes \operatorname{diag}(a)$. For necessity, transform a putative SPD factorization into the Q basis. If $\tilde{B} \otimes \tilde{A}$ is diagonal and both factors are SPD, their positive diagonal entries force every off-diagonal entry of each factor to vanish. The resulting diagonal entries factor as $S_{ij} = \tilde{A}_{ii}\tilde{B}_{jj}$. General elementwise adaptive scales fail this rank-one condition, as in SOAP (section 2).

B Theorem statements and assumptions

The bounds below follow the survey’s column-major and norm conventions, with source versions, assumptions, and parameter regimes stated for each result.

B.1 Cumulative Shampoo: the original regret certificate

Source. Gupta et al. [3], ICML 2018, Theorem 7 and its proof, PDF pp. 4–5. **Algorithm and assumptions.** Cumulative two-sided Gram matrices, exact fourth roots, convex losses, a uniform rank bound r , and Frobenius distance at most D to the comparator. A constant stepsize $D/\sqrt{2r}$ gives the regret bound in (40). Bounded iterates must be assumed or enforced by the appropriate metric projection. **Mechanism and scope.** Rank domination, monotone metric potentials, and trace concavity are derived in appendix A.1. The result excludes EMA, momentum, arbitrary grafting, intermittent roots, and finite precision. The rank bound applies to each gradient, rather than a population covariance.

B.2 One-sided structure in the unified adaptive analysis

Source. Xie et al. [19], arXiv:2503.10537v2, Theorem 4.2 and Algorithm 4, PDF p. 11 for the theorem; Theorem 4.4 gives a related convex stochastic specialization. **Algorithm.** The normalized left statistic is $L_t = \varepsilon I + (1/n) \sum_{s \leq t} G_s G_s^\top$; the variable metric is $I_n \otimes L_t^{1/2}$ in this survey’s vectorization. The update includes projection in that metric onto the specified matrix-norm domain. **Bound.** With comparator radius D_{op} , the theorem’s one-sided bound, retaining its normalization, is

$$\mathcal{R}_T \leq \left(\frac{D_{\text{op}}^2}{2n\eta} + \eta \right) \left[\sqrt{n} \operatorname{tr} \left(\left[\sum_{t=1}^T G_t G_t^\top \right]^{1/2} \right) + mn\sqrt{\varepsilon} \right]. \quad (76)$$

For a domain $\|W\|_{\text{op}} \leq R$, taking $D_{\text{op}} \leq 2R$ and $\eta = \sqrt{2/n}R$ gives the leading term $2\sqrt{2}R \operatorname{tr}[(\sum_t G_t G_t^\top)^{1/2}]$, plus the initialization contribution. Equation (42) uses this factor normalization; its constants depend on it. **Scope.** The framework requires structured matrix algebras. The set of two-sided Kronecker products is not a linear matrix algebra, so a separable implementation requires a separate check of the hypotheses.

B.3 ASGO: curvature and noise in a selected mode

Source. An et al. [18], arXiv:2503.20762v3, Theorem 3, PDF p. 5. **Algorithm.** The one-sided accumulator starts at zero and sums $G_t G_t^\top$; its metric adds εI *after* taking the square root, unlike $(S_t + \varepsilon I)^{1/2}$. The stepsize is $\eta = D_{\text{op}}$. **Assumptions and conclusion.** F is convex and smooth with respect to $\|X\|_A^2 = \text{tr}(X^\top A X)$; unbiased independent minibatch averaging reduces the single-sample matrix noise envelope by B . Finite spectral and Frobenius iterate radii yield the objective-gap bound (43). Dimension or architecture comparisons must retain the radii and $\text{tr}(A), \text{tr}(V)$. **Scope.** The guarantee applies to one-sided geometry, excluding two-sided Shampoo and polar updates. A horizon-independent radius for an unprojected trajectory requires further assumptions.

B.4 AdamW-style Shampoo with coupled parameters

Source. Li et al. [54], arXiv:2601.07326v4, Theorem 1 in Section 2 and Theorem 2 in Section 3. Theorem 1 retains the principal bound of v1's Theorem 2.1. **Algorithm.** Equation (44) with exact current factors, complementary exponents $p, q \in [1, \infty]$, $p^{-1} + q^{-1} = 1$, zero momentum/statistic initialization, and decoupled decay. **Noise and regularity.** F is lower bounded and L_F -smooth. Conditional noise $N_t = G_t - \nabla F(W_t)$ is unbiased and obeys matrix envelopes $\mathbb{E}[N_t N_t^\top \mid \mathcal{F}_{t-1}] \preceq \Sigma_L$ and $\mathbb{E}[N_t^\top N_t \mid \mathcal{F}_{t-1}] \preceq \Sigma_R$. Set $\sigma^2 = (\text{tr} \Sigma_L + \text{tr} \Sigma_R)/2$ and require the damped left and right statistics to be at least $\hat{\varepsilon} I$ in Loewner order, where $\hat{\varepsilon} \geq \varepsilon > 0$. **Schedule.** For $\gamma \in (0, 1)$, $\tau \in (0, 1]$, and $\hat{\sigma}^2 = \max\{\sigma^2, L_F \Delta / (T \gamma^2)\}$, Theorem 1 prescribes

$$1 - \theta = \sqrt{\frac{L_F \Delta}{T \hat{\sigma}^2}}, \quad \theta \leq \beta \leq \sqrt{\theta}, \quad \varepsilon = \frac{\tau \hat{\sigma}^2}{m + n}, \quad (77)$$

$$\eta = \sqrt{\frac{\hat{\varepsilon} \Delta}{4 L_F T \hat{\sigma}^2}}, \quad 0 \leq \lambda \leq \frac{1}{\sqrt{1152 \hat{\varepsilon}} T^{3/4}} \left(\frac{L_F^3 \hat{\sigma}^2}{\Delta} \right)^{1/4}. \quad (78)$$

It also bounds the initial operator norm by $\|W_1\|_{\text{op}} \leq \sqrt{\hat{\varepsilon} T \Delta / (L_F \hat{\sigma}^2)}$. **Conclusion.** The expected average nuclear-gradient bound is (45). The universal lower bound $\hat{\varepsilon} = \varepsilon$ gives a prefactor at most $127 \sqrt{(m+n)/\tau}$, making dimension and damping dependence explicit. **Scope.** The result excludes arbitrary fixed hyperparameters, cached inverse powers, finite precision, and compressed state. Any choice of $\hat{\varepsilon}$ must lower-bound the actual statistics. **General-exponent extension.** Theorem 2 permits $p, q \in (0, \infty]$, $p^{-1} + q^{-1} = 2 - \omega$ with $0 < \omega \leq 2$, but sets $\lambda = 0$. It retains the displayed θ , ε , and $\hat{\sigma}$ choices, takes any fixed $\beta \in (0, 1)$ independent of T , and uses

$$\eta = \hat{\varepsilon}^{1-\omega/2} \sqrt{\frac{\Delta}{4 L_F T \hat{\sigma}^2}}. \quad (79)$$

The power measure $\Psi_\omega(G) = \sum_i \sigma_i(G)^\omega$ is the source's $\|G\|_*^\omega$, not an entrywise power. Its average bound is $\mathcal{O}(T^{-\omega/4} + T^{-1/2})$ in the positive-noise regime, with constants depending on dimension, damping, noise, ω , and fixed β . At $\omega = 2$ the update reduces to momentum SGD. The general-exponent result permits fixed first momentum but requires zero decay.

B.5 Convergent Taylor-family NS for Muon

Source. Kim and Oh [32], arXiv:2601.19156v1, Definition 2 and Theorems 1–2, PDF pp. 5–6; empirical-polynomial limitation in Appendix H.4. **Algorithm and geometry.** The source uses $M_t = \beta M_{t-1} + G_t$ and $X_{t,0} = M_t / \max\{1, \|M_t\|_F\}$ before a finite Taylor-family inverse-square-root iteration. For order κ , its polynomial is

$$p_\kappa(\lambda) = \sum_{j=0}^{\kappa} \frac{\binom{2j}{j}}{4^j} (1 - \lambda)^j, \quad X_{k+1} = p_\kappa(X_k X_k^\top) X_k. \quad (80)$$

The proof requires the theorem’s normalization and support residual, and the contraction of this NS family. The objective satisfies operator-to-nuclear smoothness; the oracle is unbiased with bounded variance and uses minibatch averaging. **Schedule and bound.** In the nondegenerate stochastic regime, the stated scaling is $1 - \beta = \min\{\sqrt{L_{\text{op}} \Delta \bar{B}} / (\sigma \sqrt{rT}), 1\}$ and $\eta = \sqrt{(1 - \beta) \Delta / (T L_{\text{op}})}$, with the source’s momentum convention. The bound (49) includes a finite-iteration factor depending on the initialized support residual. For a uniform constant along the outer trajectory, this residual must stay uniformly below one. Exact polar removes the approximation factor. **Scope.** The empirical quintic coefficient triple $(3.4445, -4.775, 2.0315)$ lies outside this Taylor family, so its finite-step map requires a separate argument, as do weight decay, auxiliary optimizers, and finite precision.

B.6 Practical five-step Muon with decay

Source. Wang et al. [31], arXiv:2609.33152v1, Theorem 1, PDF p. 6; constants are defined in its equations (11) and (20). **Algorithm.** The empirical five-step quintic, a Nesterov-style momentum input, a normalization floor, and decoupled decay are retained in the mathematical update. Frobenius smoothness, unbiased bounded-variance minibatch gradients, and exact arithmetic are assumed. **Schedule.** With the source’s positive constants c_1, c_2, c_3, C_2, C_3 ,

$$\eta = c_1 T^{-3/4}, \quad 1 - \beta = c_2 T^{-1/2}, \quad 0 < \varepsilon \leq c_3 T^{-1/4}, \quad \lambda = \frac{C_2}{2(\|W_0\|_F + C_3 c_1 T^{1/4})}. \quad (81)$$

The decay must follow the last relation. The stochastic bound is $\mathcal{O}((L_F \Delta \sigma^2 / B)^{1/4} T^{-1/4})$ in the positive-noise regime; its constants do not cover degenerate zero-noise cases by substitution. **Scope.** The theorem analyzes the empirical polynomial under a coupled schedule in exact arithmetic. Matrix dimensions and coefficient estimates remain in the constants; fixed schedules and low precision require further analysis.

B.7 Layer-wise finite NS and Nesterov tracking

Source. Peng et al. [33], arXiv:2609.39595v1, Theorem 3.4 and Corollary 3.5, PDF pp. 8–9. **Algorithm.** At the same full parameter state, each matrix block uses its own Nesterov momentum input and fixed-depth empirical NS map, with Frobenius normalization and a separate zero-input case. The update uses exact arithmetic and *no weight decay*, and excludes auxiliary AdamW for other parameter classes. **Assumptions.** The objective is lower bounded, with the paper’s blockwise generalized smoothness and bounded-variance stochastic oracle. Blocks need not be independent,

and the objective need not be separable. Standard Frobenius smoothness is a specialization of the smoothness setting. **Schedule and conclusion.** Writing $\delta = 1 - \beta_1$, the corollary takes $\delta = \delta_0 T^{-1/2}$, $\eta = \eta_0 T^{-3/4}$, and $\beta_2 = 1 - \delta^a$ for $a > 1/2$. For fixed block count and scalar-map, noise, and regularity constants, the average sum of blockwise Frobenius gradient norms is $\mathcal{O}(T^{-1/4})$. Matching the two momentum coefficients corresponds to $a = 1$. **Mechanism and scope.** Positive alignment and bounded update energy hold for the finite map, with tracking errors controlled by the coupled schedules. No uniform exact-polar approximation is required. The estimate has no explicit dimension/rank factor, but a width-uniform claim requires control of its other constants. It gives no memory-cost bound.

B.8 Variance-reduced Muon: a different oracle

Source. Chang et al. [34], arXiv:2509.15816v6, Theorems 6 and 8, PDF pp. 9–10; additional PL results impose further assumptions. **MVR1.** The basic schedule is $\eta_t = t^{-3/4}$ and $\beta_t = 1 - t^{-1/2}$, with the specified bounded decay schedule. The expected average Frobenius-gradient rate is $\tilde{\mathcal{O}}(T^{-1/4})$, where the tilde includes factors such as $\log T$. **MVR2.** The variance-reduced schedule is $\eta_t = t^{-2/3}$ and $\beta_t = 1 - t^{-2/3}$, with the corresponding decay restriction. The rate is $\tilde{\mathcal{O}}(T^{-1/3})$ under stronger assumptions, including mean-square smoothness. The correction evaluates the *same fresh sample* at current and previous iterates, rather than reusing a stored noisy gradient. **Comparability.** Comparisons with single-evaluation Muon must count gradient evaluations and retain sample-reuse and oracle-smoothness requirements. The faster PL conclusions additionally require gradient bounds and parameter restrictions beyond nonconvex stationarity.

B.9 Soft polar and asymptotic guarantees

Source. de Oliveira et al. [52], arXiv:2609.30546v1, Theorem 1 and Corollary 1, followed by the soft-Muon specialization and boundedness analysis. **Object and assumptions.** The deterministic heavy-ball-type recursion uses uniformly bounded SPD preconditioners and restricted step/momentum parameters. The objective satisfies smoothness and properness/bounded-sublevel conditions. In the soft-polar specialization, the left matrix is $(MM^\top + \varepsilon^2 I)^{1/2}$: the positive floor gives a lower bound, while an upper bound requires momentum control. **Conclusion and scope.** The Lyapunov analysis gives asymptotic gradient convergence and a linear rate under the additional PL condition. The continuous soft-polar transform differs from a fixed empirical polynomial with Frobenius normalization; stochastic finite-precision Muon and arbitrary fixed practical schedules remain outside the result.

B.10 Generalized SOAP: rotation without covariance transport

Source. Sun et al. [13], arXiv:2609.24522v1, Algorithm 4; Theorem 1 in Section 3.3.1; proof in Appendix B. **Update.** Equation (46), $M_0 = V_0 = 0$, arbitrary current-sample-dependent orthogonal rotations; entrywise division and square root. **Schedule.** For $\gamma, \tau \in (0, 1]$, define $\hat{\sigma}^2 = \max\{\sigma^2, L_F \Delta / (T \gamma^2)\}$. Require $\sqrt{V_{t,ij} + \varepsilon} \geq \delta > 0$ and choose

$$1 - \theta = \sqrt{\frac{L_F \Delta}{T \hat{\sigma}^2}}, \quad 0 < \beta < 1, \quad \eta = \sqrt{\frac{\delta^2 \Delta}{4 L_F T \hat{\sigma}^2}}, \quad \varepsilon = \frac{\tau \hat{\sigma}^2}{m + n}. \quad (82)$$

Conclusion. Equation (47). Taking $\delta = \sqrt{\varepsilon}$ gives prefactor $(2 + 16/\sqrt{\tau})\sqrt{m+n}$. **Scope.** The result excludes decay, bias correction, transported second moments, and finite precision. The related SPlus theorem concerns truncated internal iterates, rather than its averaged output.

B.11 Two geometries: nonconvex adaptive families

Source. Xie et al. [11], arXiv:2511.20584v1, Algorithm 1, Theorems C.7–C.8; noise Assumption C.1; normalized-descent comparison in Theorems 4.6 and 4.9. **Guarantee and assumptions.** Cumulative and EMA adaptation over well-structured metric cones controls the average induced dual-gradient norm at $\tilde{\mathcal{O}}(T^{-1/4})$, including one-sided Shampoo. Besides unbiasedness, Assumption C.1 requires, almost surely,

$$-\Sigma \preceq \nabla F(x) \nabla F(x)^\top - g(x)g(x)^\top \preceq \Sigma. \quad (83)$$

This is stronger than a noise covariance envelope. The EMA specialization uses $1 - \beta = \Theta(\log d/T)$ and $\eta \asymp \sqrt{\Delta/(\Lambda_{\mathcal{H}} T \log^2 d)}$, with adaptive smoothness $\Lambda_{\mathcal{H}}$ and the structured noise trace in the constants. The NSD adaptive-noise comparison changes the noise premise; the dimension-dependent lower bound concerns sign descent. Arbitrary two-sided updates, momentum/decay, and finite NS remain outside the guarantee.

C Reference specifications and historical context

The tables expand the comparison in section 4.5, detailing recursions, state, numerical work, and distributed implementations.

C.1 Representative method specifications

Tables 6 and 7 compare representative recursions for an $m \times n$ block. Here D_t denotes the gradient-driven direction in (9); learning-rate schedules, decoupled decay, and separate block scales are outer choices. Referenced equations specify initialization, damping, and averaging.

Table 6: State, update input, and geometry. KL population targets and SOAP fixed-basis equations are distinguished from their online updates.

Method / anchor	Retained state and input	Direction and represented geometry
AdamW [2]	Entrywise EMA states M_t, V_t , fed by G_t and $G_t^{\odot 2}$	Bias-corrected diagonal scaling of momentum, (8)
Cumulative Shampoo [3]	L_t, R_t : cumulative mode Grams; input G_t	$L_t^{-1/4} G_t R_t^{-1/4}$; one separable operator, (12)
EMA Shampoo family [54]	Momentum M_t and EMA mode Grams L_t, R_t , fed by G_t	Damped one-/two-sided powers applied to M_t ; powers and decay in (44)
SOAP [5]	Mode Grams/bases and entrywise moments in $Q_L^\top G_t Q_R$	Rotated Adam scaling, (15); the basis is factored but the scales need not be

Table 6 continued

Method / anchor	Retained state and input	Direction and represented geometry
KL-Shampoo [10]	Coupled factor estimates; practical bases and factor eigenvalue scales; gradient or momentum input	Inverse-square-root factor action. Population target in (18); separate online estimation and QR recursions (source Secs. 3.2–4)
ASGO, Algorithm 1 [18]	One left accumulator $V_t^L = \sum_{s=0}^t G_s G_s^\top$; input G_t	$[(V_t^L)^{1/2} + \varepsilon I]^{-1} G_t$; damping is outside the square root
Finite Muon [6, 31]	Momentum buffer; input Z_t with a declared momentum convention	Scaled output $a_t X_K$ of (23); finite singular-value map in (24)

Table 7: Numerical and distributed implementations. State orders count dense unblocked optimizer arrays, excluding parameters, gradients, and transient workspace; constants depend on caching.

Method	State order	Numerical work / refresh	Requirement for the same logical update
AdamW	$O(mn)$	Elementwise state and scaling every step	Coordinate shards retain the corresponding moments and synchronized gradient input
Cumulative Shampoo	$O(m^2 + n^2)$	Current exact powers in the canonical recursion	Factor sharding preserves full mode Grams and their action
EMA Shampoo family	$O(m^2 + n^2 + mn)$	Current powers in the theorem; specify cached roots and blocking separately	Synchronize EMA/momentum states and preserve logical blocks
SOAP	$O(m^2 + n^2 + mn)$	Basis refresh by EVD or warm-started QR; coordinate moments each step	Preserve rotations, basis ownership, and the state rule at each basis change
KL-Shampoo with momentum	$O(m^2 + n^2 + mn)$	Coupled online estimation; QR basis refresh and eigenvalue-scale updates in the practical scheme	Preserve weighted cross-mode products and the same estimator/refresh schedule
ASGO, Algorithm 1	$O(m^2)$	Current square root and damped inverse	Preserve the full selected-mode accumulator; a row-block diagonal substitute changes it
Finite Muon	$O(mn)$	K polynomial steps with specified normalization and precision	Preserve global normalization and products; proposition 8.2 gives one exact sharded construction

Dense arithmetic costs. AdamW requires $O(mn)$ elementwise work. Forming both mode Grams and applying dense two-sided factors or bases costs $O(mn(m+n))$; full EVD or QR refreshes cost $O(m^3 + n^3)$. A left-only mode requires $O(m^2 n)$ products and a dense $O(m^3)$ decomposition.

A K -step Muon polynomial using the smaller Gram costs $O(Kmn \min(m, n))$. Execution time also depends on refresh frequency, solver depth, blocking, kernel utilization, and communication. Iterative KL fitting adds work according to its iteration schedule.

SOAP enlarges the scaling class while retaining a factored basis; a finite Muon polynomial changes the singular response while preserving singular directions in exact arithmetic. KL fitting changes what the factors estimate, while sharding can preserve the chosen operator. Each requires a different analysis: state transport, alignment/energy, estimator tracking, and distributed product fidelity, respectively.

C.2 Conceptual anchors

Table 8: Conceptual anchors for the comparison.

Anchor	Primary focus	Role in the comparison
Old Optimizer, New Norm [7]	Norm-based interpretations of instantaneous first-order updates	Separate direction, magnitude, and the effects of historical adaptation
Clarifying Shampoo [9]	Adapted spectral descent and trajectory-dependent Shampoo geometry	State conditions for adapted-polar identities and distinguish them from identical dynamics
KL-Shampoo / KL-SOAP [10]	Divergence-based factor estimation and practical basis/scale updates	Separate a population fitting target, its online estimator, and the resulting operator
A Tale of Two Geometries [11]	Adaptive versus non-Euclidean smoothness and stochastic variance	Compare rates with their geometry and oracle assumptions retained
ARO [12]	Adaptive rotation, normed descent, and matrix optimizer design	Place moving-basis adaptation and spectral transformations in a shared specification
Rotation-based convergence [13]	A unified analysis of specified SOAP-, Conda-, and SPlus-type recursions	Attach rotation-based claims to the analyzed normalizer, state, and schedules

C.3 Historical development

Table 9: Development of structured preconditioning and spectral update design.

Period	Structured adaptive preconditioning	Spectral update design and Muon
2018–2023	Original Shampoo and its cumulative regret analysis; scalable and distributed realizations [3, 4, 26]	Before Muon; norm-based updates and structured preconditioning provide distinct antecedents.
2024	Statistical reinterpretation of the factors; SOAP separates a learned basis from coordinate scaling [5, 20]	Norm-based synthesis and Muon’s momentum-to-polar construction, computed through matrix polynomials [6, 7]

Table 9 continued

Period	Structured adaptive preconditioning	Spectral update design and Muon
2025	Heuristic decomposition, stable whitening, KL fitting, and structure-sensitive convex guarantees [10, 18, 19, 21, 25]	Scaling recipes, adaptive spectral updates, and communication-aware compression [8, 40, 41, 45]
2026	Practical-form nonconvex theory, adapted-polar connections, rotation-based convergence, and solver/precision-aware refinements [9, 13, 28, 29, 54]	Taylor-NS and actual finite-map guarantees; rotation-based design, soft proxies, and curvature-aware compositions [12, 31–33, 42, 52]

D Empirical records

The six observations in [section 9](#) are listed with source versions, locators, numerical settings, and extraction status.

Table 10: Empirical observations. **NE** means *not extracted for this observation*, including fields whose applicability remains to be checked, rather than omitted by the paper. Precision refers only to the named computation.

Source / locator	Setting, budget, tuning, and numerical recipe	Metric and observed outcome
SOAP v2 [5] Fig. 1 (right), Secs. 5, 6.1–6.2; App. A	C4; 660M decoder-only model (non-embedding count); reference regime: $20\times$ model-size tokens, 2M-token batches. Learning-rate sweep, then paired sweeps of learning rate and each other hyperparameter. Refresh intervals from 1 to 100; SOAP uses initialized eigenvectors and subsequent QR-based updates. Mixed-bf16 training; factor updates in fp32.	Loss versus refresh interval. SOAP and Shampoo are close at interval 1; both degrade at longer intervals, with slower degradation for SOAP, suggesting different basis and scale adaptation timescales in this regime.
Purifying Shampoo v1 [21] Fig. 2, Sec. 3; App. E.1	Imagewoof; SimpleViT and ConvNeXt V2; 100 epochs, batch 128. Learning-rate grids differ by method/model; $\epsilon = 10^{-10}$. Grafted Shampoo and EShampoo refresh roots/bases every 100 steps, respectively; eigenvalue correction updates each step. One A100 per run. Precision: NE.	Final training cross-entropy versus learning rate. EShampoo matches or improves on Adam-grafted Shampoo for both models. Ungrafted Shampoo matches the ViT result but not ConvNeXt V2, separating scale correction from model-dependent effects.
KL-Shampoo v10 [10] Fig. 9; Sec. 2 and App. H	450M Llama3. Random search: 120 runs per method. QR every 10 iterations in fp32; refresh frequency was not tuned for runtime. Panel-specific data, token budget, batch size, other state precision, and search ranges: NE.	Progress per iteration and wall-clock time. KL-Shampoo improves on Muon per iteration, while Muon is better per time; both improve on SOAP. The ranking depends on the denominator under this schedule.
Scalable Muon v1 [8] Table 1, Sec. 3.1; recipe in Sec. 2.2	Approximately 800M dense model, with a two-layer $[H, 4H]$ MLP replacing SwiGLU. Evaluated at 4B tokens of a 20B-token schedule. Compares baseline scaling, RMS normalization, and shape-adjusted learning rate. Recipe: five Newton–Schulz steps, momentum 0.95. Panel-specific tuning budget, data, and precision: NE.	Validation loss: baseline 2.812; normalized update 2.789; adjusted learning rate 2.789, supporting shape-aware scaling within this recipe. Weight-RMS diagnostics accompany the loss.

Table 10 continued

Source / locator	Setting, budget, tuning, and numerical recipe	Metric and observed outcome
Polar Express v5 [35] Fig. 5, Secs. 4.2–4.3	GPT-2-Small (124M), FineWeb, 1B tokens; batch 32, four H100s. Fixed learning rate 0.05 and weight decay 0.1 in this ablation. Varies inner depth from 2 to 30 and includes exact SVD. Mixed-bf16 training; polynomial matrix-sign computation in bf16. SVD arithmetic: NE.	Final validation loss and training-step runtime. Five or six polynomial steps improve on two or three; further accuracy, including SVD, gives no loss improvement. SVD roughly doubles training-step runtime. The denominator includes forward/backward computation.
DASH v1 [29] Table 1, Sec. 5	Llama-953M, C4; 9,089 steps, 2M-token batch, sequence length 1,024; three seeds. AdamW learning rate selected from six candidates, then transferred through Adam grafting. Matched row: block 2,048, coupled Newton inverse-root solver, refresh every step, Frobenius normalization, fp32 inverse roots.	Mean validation perplexity and <i>optimizer-step</i> latency. DIST and DASH both give perplexity 11.87; latency is 675 versus 221 ms, excluding forward/backward passes.